Maximum Likelihood Estimation

From an economic model to prediction errors and estimates

Gauthier Vermandel

Paris-Dauphine – PSL

26 November 2026

Why estimate a solved model?

Last session

The production economy delivered a decision rule: a productivity shock propagates through consumption, capital and output.

The economic question

Which parameter values make that mechanism plausible for an observed sample? Matching a few moments does not use all the information in the data.

Today

Map model variables to observables, use a filter to compute prediction errors, then construct and inspect the likelihood before maximizing it.

Why estimate a solved model?

Last session

The production economy delivered a decision rule: a productivity shock propagates through consumption, capital and output.

The economic question

Which parameter values make that mechanism plausible for an observed sample? Matching a few moments does not use all the information in the data.

Today

Map model variables to observables, use a filter to compute prediction errors, then construct and inspect the likelihood before maximizing it.

Why estimate a solved model?

Last session

The production economy delivered a decision rule: a productivity shock propagates through consumption, capital and output.

The economic question

Which parameter values make that mechanism plausible for an observed sample? Matching a few moments does not use all the information in the data.

Today

Map model variables to observables, use a filter to compute prediction errors, then construct and inspect the likelihood before maximizing it.

Plan

1. Introduction
2. Filters
3. Maximum likelihood
4. Estimation with Dynare

Why use the sample rather than only selected moments to discipline parameters?

How we use the session

Teaching block	Effective minutes
Model, measurement and filters	60
Likelihood and inversion benchmark	35
Data and Dynare estimation	50
Handout checkpoints and synthesis	35
Teaching time	180
Separate break after the likelihood block	15

Read the handout explanations alongside the derivations. Exercises remain in the handout and extend the live walkthrough.

Calibration, moments and full information

- Calibration: external evidence and selected moments fix parameter values in toy models built to display simple mechanisms.
- Matching moments: a limited-information (weak) method that targets chosen statistics of the data.
- Estimation: a full-information method that uses the entire sample, by maximum likelihood or Bayesian inference.

The calibration critique motivates the likelihood approach: *"Because the modeler has so much freedom in choosing functional forms, parameter values, and other inputs, the model can be used to obtain almost any result one desires"*.[1]

[1] Pindyck (2017), *The Use and Misuse of Models for Climate Policy*, quoted in handout §1. Kydland and Prescott (1982) calibrated their prototype to moments, Taylor and Uhlig (1990) and Fair and Taylor (1990) are early full-information examples.

Calibration, moments and full information

- Calibration: external evidence and selected moments fix parameter values in toy models built to display simple mechanisms.
- Matching moments: a limited-information (weak) method that targets chosen statistics of the data.
- Estimation: a full-information method that uses the entire sample, by maximum likelihood or Bayesian inference.

The calibration critique motivates the likelihood approach: *"Because the modeler has so much freedom in choosing functional forms, parameter values, and other inputs, the model can be used to obtain almost any result one desires"*.[1]

[1] Pindyck (2017), *The Use and Misuse of Models for Climate Policy*, quoted in handout §1. Kydland and Prescott (1982) calibrated their prototype to moments, Taylor and Uhlig (1990) and Fair and Taylor (1990) are early full-information examples.

Calibration, moments and full information

- Calibration: external evidence and selected moments fix parameter values in toy models built to display simple mechanisms.
- Matching moments: a limited-information (weak) method that targets chosen statistics of the data.
- Estimation: a full-information method that uses the entire sample, by maximum likelihood or Bayesian inference.

The calibration critique motivates the likelihood approach: *"Because the modeler has so much freedom in choosing functional forms, parameter values, and other inputs, the model can be used to obtain almost any result one desires"*.[1]

[1] Pindyck (2017), *The Use and Misuse of Models for Climate Policy*, quoted in handout §1. Kydland and Prescott (1982) calibrated their prototype to moments, Taylor and Uhlig (1990) and Fair and Taylor (1990) are early full-information examples.

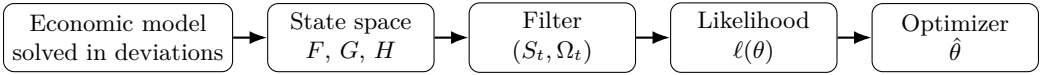
Calibration, moments and full information

- Calibration: external evidence and selected moments fix parameter values in toy models built to display simple mechanisms.
- Matching moments: a limited-information (weak) method that targets chosen statistics of the data.
- Estimation: a full-information method that uses the entire sample, by maximum likelihood or Bayesian inference.

The calibration critique motivates the likelihood approach: *"Because the modeler has so much freedom in choosing functional forms, parameter values, and other inputs, the model can be used to obtain almost any result one desires"*.[1]

[1] Pindyck (2017), *The Use and Misuse of Models for Climate Policy*, quoted in handout §1. Kydland and Prescott (1982) calibrated their prototype to moments, Taylor and Uhlig (1990) and Fair and Taylor (1990) are early full-information examples.

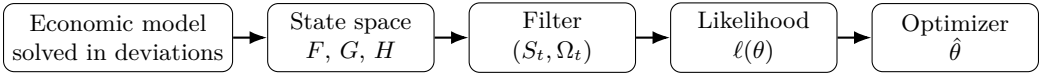
From the model to the estimate



- Filters recover the sequence of shocks that a model needs to replicate a sample.
- For a given sample, infinitely many shock sequences and models can replicate the data: each extracted sequence is one representation among many.[2]
- Every candidate θ requires re-solving the model and re-running the filter: the criterion is only meaningful after the observables are declared.

[2] Handout §2. Estimation therefore picks the parameters whose implied joint density is most favored by the data.

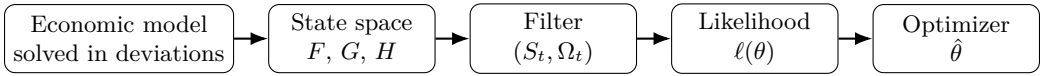
From the model to the estimate



- Filters recover the sequence of shocks that a model needs to replicate a sample.
- For a given sample, infinitely many shock sequences and models can replicate the data: each extracted sequence is one representation among many.[2]
- Every candidate θ requires re-solving the model and re-running the filter: the criterion is only meaningful after the observables are declared.

[2] Handout §2. Estimation therefore picks the parameters whose implied joint density is most favored by the data.

From the model to the estimate



- Filters recover the sequence of shocks that a model needs to replicate a sample.
- For a given sample, infinitely many shock sequences and models can replicate the data: each extracted sequence is one representation among many.[2]
- Every candidate θ requires re-solving the model and re-running the filter: the criterion is only meaningful after the observables are declared.

[2] Handout §2. Estimation therefore picks the parameters whose implied joint density is most favored by the data.

Plan

1. Introduction
2. Filters
3. Maximum likelihood
4. Estimation with Dynare

A solution generates data. A filter asks which hidden states can explain the observations.

The production economy from Session 2

$$\bar{k} = \left[\frac{\beta^{-1} - (1-\delta)}{\alpha}\right]^{1/(\alpha-1)}, \qquad \bar{c} = \bar{k}^{\alpha} - \delta\bar{k}$$

$$\hat{k}_t = \bar{k}^{\alpha-1}\hat{a}_t + \left[\alpha\bar{k}^{\alpha-1} + 1 - \delta\right]\hat{k}_{t-1} - \frac{\bar{c}}{\bar{k}}\hat{c}_t$$

$$\hat{c}_t = E_t\hat{c}_{t+1} - \beta\alpha\bar{k}^{\alpha-1}\left[E_t\hat{a}_{t+1} + (\alpha-1)\hat{k}_t\right]$$

$$\hat{a}_t = \rho_A\hat{a}_{t-1} + \varepsilon_t$$

- Hats are log deviations from the steady state, capital used in production is predetermined.
- Fixed labour, one technology shock, calibration $\alpha = 0.40$, $\beta = 0.99$, $\delta = 0.025$.[3]
- When output is observed, the production relation $\hat{y}_t = \hat{a}_t + \alpha\hat{k}_{t-1}$ is added to the system.

[3] Teaching scripts `kalmanRBC.m` and `invertRBC.m`, session 3 code.

The production economy from Session 2

$$\bar{k} = \left[\frac{\beta^{-1} - (1-\delta)}{\alpha}\right]^{1/(\alpha-1)}, \qquad \bar{c} = \bar{k}^{\alpha} - \delta\bar{k}$$

$$\hat{k}_t = \bar{k}^{\alpha-1}\hat{a}_t + \left[\alpha\bar{k}^{\alpha-1} + 1 - \delta\right]\hat{k}_{t-1} - \frac{\bar{c}}{\bar{k}}\hat{c}_t$$

$$\hat{c}_t = E_t\hat{c}_{t+1} - \beta\alpha\bar{k}^{\alpha-1}\left[E_t\hat{a}_{t+1} + (\alpha-1)\hat{k}_t\right]$$

$$\hat{a}_t = \rho_A\hat{a}_{t-1} + \varepsilon_t$$

- Hats are log deviations from the steady state, capital used in production is predetermined.
- Fixed labour, one technology shock, calibration $\alpha = 0.40$, $\beta = 0.99$, $\delta = 0.025$.[3]
- When output is observed, the production relation $\hat{y}_t = \hat{a}_t + \alpha\hat{k}_{t-1}$ is added to the system.

[3] Teaching scripts `kalmanRBC.m` and `invertRBC.m`, session 3 code.

The production economy from Session 2

$$\bar{k} = \left[\frac{\beta^{-1} - (1-\delta)}{\alpha}\right]^{1/(\alpha-1)}, \qquad \bar{c} = \bar{k}^{\alpha} - \delta\bar{k}$$

$$\hat{k}_t = \bar{k}^{\alpha-1}\hat{a}_t + \left[\alpha\bar{k}^{\alpha-1} + 1 - \delta\right]\hat{k}_{t-1} - \frac{\bar{c}}{\bar{k}}\hat{c}_t$$

$$\hat{c}_t = E_t\hat{c}_{t+1} - \beta\alpha\bar{k}^{\alpha-1}\left[E_t\hat{a}_{t+1} + (\alpha-1)\hat{k}_t\right]$$

$$\hat{a}_t = \rho_A\hat{a}_{t-1} + \varepsilon_t$$

- Hats are log deviations from the steady state, capital used in production is predetermined.
- Fixed labour, one technology shock, calibration $\alpha = 0.40$, $\beta = 0.99$, $\delta = 0.025$.[3]
- When output is observed, the production relation $\hat{y}_t = \hat{a}_t + \alpha\hat{k}_{t-1}$ is added to the system.

[3] Teaching scripts `kalmanRBC.m` and `invertRBC.m`, session 3 code.

The state-space representation

Definition: state-space representation

$$x_t = Fx_{t-1} + G\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$
$$y_t = Hx_t + v_t, \qquad v_t \sim \mathcal{N}(0, R), \qquad E(\varepsilon_t v_t') = 0.$$

The first equation is the solved policy rule. The second maps latent model variables to the data. F need not have full rank.

$$y_t = HFx_{t-1} + HG\varepsilon_t + v_t.$$

The observation combines the inherited state, today's structural innovation and measurement noise. These are handout equations (1)–(3).

The state-space representation

Definition: state-space representation

$$x_t = Fx_{t-1} + G\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$
$$y_t = Hx_t + v_t, \qquad v_t \sim \mathcal{N}(0, R), \qquad E(\varepsilon_t v_t') = 0.$$

The first equation is the solved policy rule. The second maps latent model variables to the data. F need not have full rank.

$$y_t = HFx_{t-1} + HG\varepsilon_t + v_t.$$

The observation combines the inherited state, today's structural innovation and measurement noise. These are handout equations (1)–(3).

The state-space representation

Definition: state-space representation

$$x_t = Fx_{t-1} + G\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$
$$y_t = Hx_t + v_t, \qquad v_t \sim \mathcal{N}(0, R), \qquad E(\varepsilon_t v_t') = 0.$$

The first equation is the solved policy rule. The second maps latent model variables to the data. F need not have full rank.

$$y_t = HFx_{t-1} + HG\varepsilon_t + v_t.$$

The observation combines the inherited state, today's structural innovation and measurement noise. These are handout equations (1)–(3).

The state-space representation

Definition: state-space representation

$$x_t = Fx_{t-1} + G\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$
$$y_t = Hx_t + v_t, \qquad v_t \sim \mathcal{N}(0, R), \qquad E(\varepsilon_t v_t') = 0.$$

The first equation is the solved policy rule. The second maps latent model variables to the data. F need not have full rank.

$$y_t = HFx_{t-1} + HG\varepsilon_t + v_t.$$

The observation combines the inherited state, today's structural innovation and measurement noise. These are handout equations (1)–(3).

An RBC state-space example

For the artificial consumption sample, use the Session 2 policy coefficients:

$$x_t = \begin{bmatrix} \hat{k}_t \\ \hat{a}_t \\ \hat{c}_t \end{bmatrix}, \quad F = \begin{bmatrix} F_k & G_k\rho_A & 0 \\ 0 & \rho_A & 0 \\ F_c & G_c\rho_A & 0 \end{bmatrix}, \quad G = \begin{bmatrix} G_k \\ 1 \\ G_c \end{bmatrix}.$$

$$H = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}, \qquad y_t = \hat{c}_t, \qquad R = 0.$$

- F and G come from the economic solution, while H states what is observed.
- One consumption series informs three model variables. Capital and productivity are latent.
- The script `kalmanRBC.m` implements this observation choice, not the GDP observation used later.

An RBC state-space example

For the artificial consumption sample, use the Session 2 policy coefficients:

$$x_t = \begin{bmatrix} \hat{k}_t \\ \hat{a}_t \\ \hat{c}_t \end{bmatrix}, \quad F = \begin{bmatrix} F_k & G_k\rho_A & 0 \\ 0 & \rho_A & 0 \\ F_c & G_c\rho_A & 0 \end{bmatrix}, \quad G = \begin{bmatrix} G_k \\ 1 \\ G_c \end{bmatrix}.$$

$$H = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}, \qquad y_t = \hat{c}_t, \qquad R = 0.$$

- F and G come from the economic solution, while H states what is observed.
- One consumption series informs three model variables. Capital and productivity are latent.
- The script `kalmanRBC.m` implements this observation choice, not the GDP observation used later.

An RBC state-space example

For the artificial consumption sample, use the Session 2 policy coefficients:

$$x_t = \begin{bmatrix} \hat{k}_t \\ \hat{a}_t \\ \hat{c}_t \end{bmatrix}, \quad F = \begin{bmatrix} F_k & G_k\rho_A & 0 \\ 0 & \rho_A & 0 \\ F_c & G_c\rho_A & 0 \end{bmatrix}, \quad G = \begin{bmatrix} G_k \\ 1 \\ G_c \end{bmatrix}.$$

$$H = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}, \qquad y_t = \hat{c}_t, \qquad R = 0.$$

- F and G come from the economic solution, while H states what is observed.
- One consumption series informs three model variables. Capital and productivity are latent.
- The script `kalmanRBC.m` implements this observation choice, not the GDP observation used later.

The measurement equation is a modeling choice

$$y_t = H x_t + v_t$$

- In the artificial exercise the observable is consumption: $H = [0\ 0\ 1]$ selects the third state, and the teaching filter sets $R = 0$.[4]
- With output observed, use $\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}$. A contemporary measurement matrix must respect the lag on capital, by an appropriate state representation.
- Declaring the observables, their units and their trend is part of the statistical model: changing them changes the likelihood, not only the data.

[4] `kalmanRBC.m`: `y_obs = H*x_` with `R = 0`. Capital and productivity are treated as unobserved.

The measurement equation is a modeling choice

$$y_t = H x_t + v_t$$

- In the artificial exercise the observable is consumption: $H = [0\ 0\ 1]$ selects the third state, and the teaching filter sets $R = 0$.[4]
- With output observed, use $\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}$. A contemporary measurement matrix must respect the lag on capital, by an appropriate state representation.
- Declaring the observables, their units and their trend is part of the statistical model: changing them changes the likelihood, not only the data.

[4] `kalmanRBC.m`: `y_obs = H*x_` with `R = 0`. Capital and productivity are treated as unobserved.

The measurement equation is a modeling choice

$$y_t = H x_t + v_t$$

- In the artificial exercise the observable is consumption: $H = [0\ 0\ 1]$ selects the third state, and the teaching filter sets $R = 0$.[4]
- With output observed, use $\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}$. A contemporary measurement matrix must respect the lag on capital, by an appropriate state representation.
- Declaring the observables, their units and their trend is part of the statistical model: changing them changes the likelihood, not only the data.

[4] `kalmanRBC.m`: `y_obs = H*x_` with `R = 0`. Capital and productivity are treated as unobserved.

Three information sets, three questions

$$\underbrace{x_{t|t-1} = E(x_t \mid Y_{1:t-1})}_{\text{prediction}} \quad \longrightarrow \quad \underbrace{E(x_t \mid Y_{1:t})}_{\text{filtering}} \quad \longrightarrow \quad \underbrace{E(x_t \mid Y_{1:T})}_{\text{smoothing}}$$

- Prediction: what did the model expect before seeing y_t?[5]
- Filtering: what does today's observation teach us about today's state?
- Smoothing: what does the whole sample teach us about past states and shocks?

The revised handout writes $x_{t|t}$ for the filtered mean and $P_{t|t}$ for its covariance. $P_{t|t-1}$ is the prediction-error covariance. These are estimates of x_t, not the latent variable itself.

[5] $E_{t-1}\{\cdot\}$ conditions on observations up to date $t-1$ (handout §2.1).

Three information sets, three questions

$$\underbrace{x_{t|t-1} = E(x_t \mid Y_{1:t-1})}_{\text{prediction}} \longrightarrow \underbrace{E(x_t \mid Y_{1:t})}_{\text{filtering}} \longrightarrow \underbrace{E(x_t \mid Y_{1:T})}_{\text{smoothing}}$$

- Prediction: what did the model expect before seeing y_t?[5]
- Filtering: what does today's observation teach us about today's state?
- Smoothing: what does the whole sample teach us about past states and shocks?

The revised handout writes $x_{t|t}$ for the filtered mean and $P_{t|t}$ for its covariance. $P_{t|t-1}$ is the prediction-error covariance. These are estimates of x_t, not the latent variable itself.

[5] $E_{t-1}\{\cdot\}$ conditions on observations up to date $t-1$ (handout §2.1).

Three information sets, three questions

$$\underbrace{x_{t|t-1} = E(x_t \mid Y_{1:t-1})}_{\text{prediction}} \longrightarrow \underbrace{E(x_t \mid Y_{1:t})}_{\text{filtering}} \longrightarrow \underbrace{E(x_t \mid Y_{1:T})}_{\text{smoothing}}$$

- Prediction: what did the model expect before seeing y_t?[5]
- Filtering: what does today's observation teach us about today's state?
- Smoothing: what does the whole sample teach us about past states and shocks?

The revised handout writes $x_{t|t}$ for the filtered mean and $P_{t|t}$ for its covariance. $P_{t|t-1}$ is the prediction-error covariance. These are estimates of x_t, not the latent variable itself.

[5] $E_{t-1}\{\cdot\}$ conditions on observations up to date $t-1$ (handout §2.1).

Three information sets, three questions

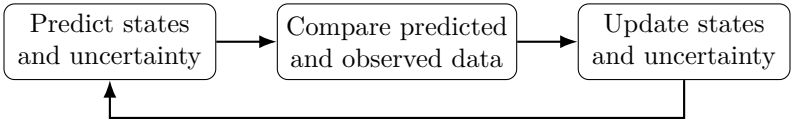
$$\underbrace{x_{t|t-1} = E(x_t \mid Y_{1:t-1})}_{\text{prediction}} \quad \longrightarrow \quad \underbrace{E(x_t \mid Y_{1:t})}_{\text{filtering}} \quad \longrightarrow \quad \underbrace{E(x_t \mid Y_{1:T})}_{\text{smoothing}}$$

- Prediction: what did the model expect before seeing y_t?[5]
- Filtering: what does today's observation teach us about today's state?
- Smoothing: what does the whole sample teach us about past states and shocks?

The revised handout writes $x_{t|t}$ for the filtered mean and $P_{t|t}$ for its covariance. $P_{t|t-1}$ is the prediction-error covariance. These are estimates of x_t, not the latent variable itself.

[5] $E_{t-1}\{\cdot\}$ conditions on observations up to date $t-1$ (handout §2.1).

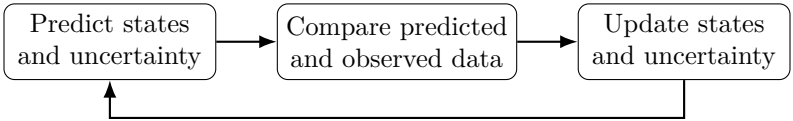
Prediction, surprise, correction



- Take the parameters as known while running one filter pass.[6]
- Store the prediction error and its covariance at every date.
- The optimizer repeats the entire pass at each new parameter value.

[6] Handout §2.1: provide H, F, G, R, Q and the data, the filter returns S_t, Ω_t, $x_{t|t}$, $P_{t|t}$ and the forecasts.

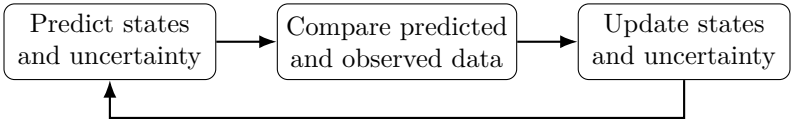
Prediction, surprise, correction



- Take the parameters as known while running one filter pass.[6]
- Store the prediction error and its covariance at every date.
- The optimizer repeats the entire pass at each new parameter value.

[6] Handout §2.1: provide H, F, G, R, Q and the data, the filter returns S_t, Ω_t, $x_{t|t}$, $P_{t|t}$ and the forecasts.

Prediction, surprise, correction



- Take the parameters as known while running one filter pass.[6]
- Store the prediction error and its covariance at every date.
- The optimizer repeats the entire pass at each new parameter value.

[6] Handout §2.1: provide H, F, G, R, Q and the data, the filter returns S_t, Ω_t, $x_{t|t}$, $P_{t|t}$ and the forecasts.

Step 1: measure the surprise

$$S_t = y_t - Hx_{t|t-1} \tag{KF1}$$

$$\Omega_t = HP_{t|t-1}H' + R \tag{KF2}$$

- S_t is the one-step forecast error of the data, not the structural shock itself.
- Ω_t is the model's own uncertainty about that error before it occurs.
- A large error is less surprising when the model predicted high uncertainty: surprises are judged relative to Ω_t.

Step 1: measure the surprise

$$S_t = y_t - Hx_{t|t-1} \tag{KF1}$$

$$\Omega_t = HP_{t|t-1}H' + R \tag{KF2}$$

- S_t is the one-step forecast error of the data, not the structural shock itself.
- Ω_t is the model's own uncertainty about that error before it occurs.
- A large error is less surprising when the model predicted high uncertainty: surprises are judged relative to Ω_t.

Step 1: measure the surprise

$$S_t = y_t - Hx_{t|t-1} \tag{KF1}$$

$$\Omega_t = HP_{t|t-1}H' + R \tag{KF2}$$

- S_t is the one-step forecast error of the data, not the structural shock itself.
- Ω_t is the model's own uncertainty about that error before it occurs.
- A large error is less surprising when the model predicted high uncertainty: surprises are judged relative to Ω_t.

Why predict with a conditional mean?

For a prediction a based only on past observations, minimize

$$E_{t-1}[(y_t - a)'(y_t - a)].$$

$$\frac{\partial}{\partial a} E_{t-1}[(y_t - a)'(y_t - a)] = -2E_{t-1}[y_t - a] = 0,$$

$$a^* = E_{t-1}[y_t] = HE_{t-1}[x_t].$$

Why this is a minimum

The error around any other predictor adds the nonnegative term $(a - a^*)'(a - a^*)$ to the minimum MSE. The cross term vanishes because $E_{t-1}[y_t - a^*] = 0$.

Why predict with a conditional mean?

For a prediction a based only on past observations, minimize

$$E_{t-1}[(y_t - a)'(y_t - a)].$$

$$\frac{\partial}{\partial a} E_{t-1}[(y_t - a)'(y_t - a)] = -2E_{t-1}[y_t - a] = 0,$$
$$a^* = E_{t-1}[y_t] = HE_{t-1}[x_t].$$

Why this is a minimum

The error around any other predictor adds the nonnegative term $(a - a^*)'(a - a^*)$ to the minimum MSE. The cross term vanishes because $E_{t-1}[y_t - a^*] = 0$.

Why predict with a conditional mean?

For a prediction a based only on past observations, minimize

$$E_{t-1}[(y_t - a)'(y_t - a)].$$

$$\frac{\partial}{\partial a} E_{t-1}[(y_t - a)'(y_t - a)] = -2E_{t-1}[y_t - a] = 0,$$
$$a^* = E_{t-1}[y_t] = HE_{t-1}[x_t].$$

Why this is a minimum

The error around any other predictor adds the nonnegative term $(a - a^*)'(a - a^*)$ to the minimum MSE. The cross term vanishes because $E_{t-1}[y_t - a^*] = 0$.

Step 2: choose how much to learn

Definition: Kalman gain

$$K_t = P_{t|t-1} H' \Omega_t^{-1} \tag{KF3}$$

- The gain scales the surprise before it moves the state: state–observation covariance relative to forecast uncertainty.
- With one state and $H = 1$, $K_t = \dfrac{P_{t|t-1}}{P_{t|t-1} + R}$: a weight in $[0, 1]$ (strictly interior when both variances are positive).
- More measurement noise lowers the weight on the new observation.
- More prior uncertainty about the state raises it.

Step 2: choose how much to learn

Definition: Kalman gain

$$K_t = P_{t|t-1} H' \Omega_t^{-1} \tag{KF3}$$

- The gain scales the surprise before it moves the state: state–observation covariance relative to forecast uncertainty.
- With one state and $H = 1$, $K_t = \dfrac{P_{t|t-1}}{P_{t|t-1} + R}$: a weight in $[0, 1]$ (strictly interior when both variances are positive).
- More measurement noise lowers the weight on the new observation.
- More prior uncertainty about the state raises it.

Step 2: choose how much to learn

Definition: Kalman gain

$$K_t = P_{t|t-1} H' \Omega_t^{-1} \tag{KF3}$$

- The gain scales the surprise before it moves the state: state–observation covariance relative to forecast uncertainty.
- With one state and $H = 1$, $K_t = \dfrac{P_{t|t-1}}{P_{t|t-1} + R}$: a weight in $[0, 1]$ (strictly interior when both variances are positive).
- More measurement noise lowers the weight on the new observation.
- More prior uncertainty about the state raises it.

Step 2: choose how much to learn

Definition: Kalman gain

$$K_t = P_{t|t-1} H' \Omega_t^{-1} \tag{KF3}$$

- The gain scales the surprise before it moves the state: state–observation covariance relative to forecast uncertainty.
- With one state and $H = 1$, $K_t = \dfrac{P_{t|t-1}}{P_{t|t-1} + R}$: a weight in $[0, 1]$ (strictly interior when both variances are positive).
- More measurement noise lowers the weight on the new observation.
- More prior uncertainty about the state raises it.

Update, then forecast one step ahead

$$x_{t|t} = x_{t|t-1} + K_t S_t \tag{KF4}$$

$$P_{t|t} = P_{t|t-1} - K_t H P_{t|t-1} \tag{KF5}$$

$$x_{t+1|t} = F x_{t|t}, \qquad P_{t+1|t} = F P_{t|t} F' + GQG' \tag{KF6–KF7}$$

- The filtered state is the prediction plus a surprise-weighted correction.
- The covariance falls in the directions the observation informs: with three states and one observable, capital and productivity remain uncertain.
- Forecasting propagates past uncertainty through F and injects new uncertainty GQG': the recursion restarts at Step 1.

Update, then forecast one step ahead

$$x_{t|t} = x_{t|t-1} + K_t S_t \tag{KF4}$$

$$P_{t|t} = P_{t|t-1} - K_t H P_{t|t-1} \tag{KF5}$$

$$x_{t+1|t} = F x_{t|t}, \qquad P_{t+1|t} = F P_{t|t} F' + GQG' \tag{KF6–KF7}$$

- The filtered state is the prediction plus a surprise-weighted correction.
- The covariance falls in the directions the observation informs: with three states and one observable, capital and productivity remain uncertain.
- Forecasting propagates past uncertainty through F and injects new uncertainty GQG': the recursion restarts at Step 1.

Update, then forecast one step ahead

$$x_{t|t} = x_{t|t-1} + K_t S_t \tag{KF4}$$

$$P_{t|t} = P_{t|t-1} - K_t H P_{t|t-1} \tag{KF5}$$

$$x_{t+1|t} = F x_{t|t}, \qquad P_{t+1|t} = F P_{t|t} F' + GQG' \tag{KF6–KF7}$$

- The filtered state is the prediction plus a surprise-weighted correction.
- The covariance falls in the directions the observation informs: with three states and one observable, capital and productivity remain uncertain.
- Forecasting propagates past uncertainty through F and injects new uncertainty GQG': the recursion restarts at Step 1.

Initialization is part of the likelihood

$$a_1 \equiv x_{1|0}, \qquad C_1 \equiv P_{1|0}$$

- The recursions need starting values $x_{1|0}$ and $P_{1|0}$ (handout §2.1).
- Known initial state x_0: $x_{1|0} = Fx_0$ and $P_{1|0} = GQG'$.
- Stable stationary model: solve the fixed point $P = FPF' + GQG'$.
- The teaching script passes $x_{1|0} = 0$ and $P_{1|0} = I$.[7]

[7] `kalmanRBC.m`: `x0 = zeros(3,1)`, `Esig0 = eye(3)`. Early likelihood terms depend on this choice.

Initialization is part of the likelihood

$$a_1 \equiv x_{1|0}, \qquad C_1 \equiv P_{1|0}$$

- The recursions need starting values $x_{1|0}$ and $P_{1|0}$ (handout §2.1).
- Known initial state x_0: $x_{1|0} = Fx_0$ and $P_{1|0} = GQG'$.
- Stable stationary model: solve the fixed point $P = FPF' + GQG'$.
- The teaching script passes $x_{1|0} = 0$ and $P_{1|0} = I$.[7]

[7] `kalmanRBC.m`: `x0 = zeros(3,1)`, `Esig0 = eye(3)`. Early likelihood terms depend on this choice.

Initialization is part of the likelihood

$$a_1 \equiv x_{1|0}, \qquad C_1 \equiv P_{1|0}$$

- The recursions need starting values $x_{1|0}$ and $P_{1|0}$ (handout §2.1).
- Known initial state x_0: $x_{1|0} = Fx_0$ and $P_{1|0} = GQG'$.
- Stable stationary model: solve the fixed point $P = FPF' + GQG'$.
- The teaching script passes $x_{1|0} = 0$ and $P_{1|0} = I$.[7]

[7] `kalmanRBC.m`: `x0 = zeros(3,1)`, `Esig0 = eye(3)`. Early likelihood terms depend on this choice.

Initialization is part of the likelihood

$$a_1 \equiv x_{1|0}, \qquad C_1 \equiv P_{1|0}$$

- The recursions need starting values $x_{1|0}$ and $P_{1|0}$ (handout §2.1).
- Known initial state x_0: $x_{1|0} = Fx_0$ and $P_{1|0} = GQG'$.
- Stable stationary model: solve the fixed point $P = FPF' + GQG'$.
- The teaching script passes $x_{1|0} = 0$ and $P_{1|0} = I$.[7]

[7] `kalmanRBC.m`: `x0 = zeros(3,1)`, `Esig0 = eye(3)`. Early likelihood terms depend on this choice.

Mini-exercise: one observation arrives

$$x_{t|t-1} = 0, \qquad P_{t|t-1} = 4, \qquad H = 1, \qquad R = 1, \qquad y_t = 3$$

- Compute S_t, Ω_t, then K_t.
- Find the filtered mean $x_{t|t}$ and covariance $P_{t|t}$.
- What changes if measurement becomes much noisier, with $R = 16$?

Work in pairs: explain the weight on the observation before calculating anything.

Mini-exercise: one observation arrives

$$x_{t|t-1} = 0, \qquad P_{t|t-1} = 4, \qquad H = 1, \qquad R = 1, \qquad y_t = 3$$

- Compute S_t, Ω_t, then K_t.
- Find the filtered mean $x_{t|t}$ and covariance $P_{t|t}$.
- What changes if measurement becomes much noisier, with $R = 16$?

Work in pairs: explain the weight on the observation before calculating anything.

Mini-exercise: one observation arrives

$$x_{t|t-1} = 0, \qquad P_{t|t-1} = 4, \qquad H = 1, \qquad R = 1, \qquad y_t = 3$$

- Compute S_t, Ω_t, then K_t.
- Find the filtered mean $x_{t|t}$ and covariance $P_{t|t}$.
- What changes if measurement becomes much noisier, with $R = 16$?

Work in pairs: explain the weight on the observation before calculating anything.

Mini-exercise: one observation arrives

$$x_{t|t-1} = 0, \qquad P_{t|t-1} = 4, \qquad H = 1, \qquad R = 1, \qquad y_t = 3$$

- Compute S_t, Ω_t, then K_t.
- Find the filtered mean $x_{t|t}$ and covariance $P_{t|t}$.
- What changes if measurement becomes much noisier, with $R = 16$?

Work in pairs: explain the weight on the observation before calculating anything.

Noisy data receive less weight

R	Ω_t	K_t	$x_{t\|t}$	$P_{t\|t}$
1	5	0.8	2.4	0.8
16	20	0.2	0.6	3.2

- The surprise is $S_t = 3$ in both cases: the data ask the same question.
- With reliable data the filter moves most of the way from 0 to 3.
- With noisy data it learns less and keeps more uncertainty: filtering weighs evidence, it does not obey it.

Noisy data receive less weight

R	Ω_t	K_t	$x_{t\|t}$	$P_{t\|t}$
1	5	0.8	2.4	0.8
16	20	0.2	0.6	3.2

- The surprise is $S_t = 3$ in both cases: the data ask the same question.
- With reliable data the filter moves most of the way from 0 to 3.
- With noisy data it learns less and keeps more uncertainty: filtering weighs evidence, it does not obey it.

Noisy data receive less weight

R	Ω_t	K_t	$x_{t\|t}$	$P_{t\|t}$
1	5	0.8	2.4	0.8
16	20	0.2	0.6	3.2

- The surprise is $S_t = 3$ in both cases: the data ask the same question.
- With reliable data the filter moves most of the way from 0 to 3.
- With noisy data it learns less and keeps more uncertainty: filtering weighs evidence, it does not obey it.

Read the recursion in MATLAB

```
Omega(t) = H*ESigma(:,:,t)*H' + R;       % (KF2)
S(t)     = Y(t) - H*Ex_hat(:,t);         % (KF1)
K        = ESigma(:,:,t)*H'/Omega(t);    % (KF3)
x_hat(:,t)   = Ex_hat(:,t) + K*S(t);     % (KF4)
Sigma(:,:,t) = ESigma(:,:,t) - K*H*ESigma(:,:,t);  % (KF5)
Ex_hat(:,t+1)  = F*x_hat(:,t);           % (KF6)
ESigma(:,:,t+1)= F*Sigma(:,:,t)*F' + G*Q*G';       % (KF7)
```

- `Ex_hat` stores $x_{t|t-1}$ and `ESigma` stores $P_{t|t-1}$: the listing mirrors the handout equations.
- The teaching implementation accepts one finite observable vector.[8]

[8] `KF.m` opens with an assertion on Y. Multi-observable filters follow the same seven equations with matrix Ω_t.

Read the recursion in MATLAB

```
Omega(t) = H*ESigma(:,:,t)*H' + R;       % (KF2)
S(t)     = Y(t) - H*Ex_hat(:,t);         % (KF1)
K        = ESigma(:,:,t)*H'/Omega(t);    % (KF3)
x_hat(:,t)   = Ex_hat(:,t) + K*S(t);     % (KF4)
Sigma(:,:,t) = ESigma(:,:,t) - K*H*ESigma(:,:,t);  % (KF5)
Ex_hat(:,t+1)  = F*x_hat(:,t);           % (KF6)
ESigma(:,:,t+1)= F*Sigma(:,:,t)*F' + G*Q*G';       % (KF7)
```

- `Ex_hat` stores $x_{t|t-1}$ and `ESigma` stores $P_{t|t-1}$: the listing mirrors the handout equations.
- The teaching implementation accepts one finite observable vector.[8]

[8] `KF.m` opens with an assertion on Y. Multi-observable filters follow the same seven equations with matrix Ω_t.

Inversion solves directly for the shock

$$Y_t = y_t = Hx_t = HFx_{t-1} + HG\varepsilon_t$$

$$\varepsilon_t = (HG)^{-1}\left(Y_t - HFx_{t-1}\right), \qquad x_t = Fx_{t-1} + G\varepsilon_t$$

- The forecast error $Y_t - HFx_{t-1}$ is the economic surprise, rescaled to shock units through $(HG)^{-1}$.[9]
- The recovered shock then generates the next state: a conditional recursion, not a covariance update.
- This displayed inverse is the linear case. Nonlinear inversion solves the corresponding nonlinear observation relation. Both require admissible initialization ($x_0 = 0$ here).

[9] Inversion filter of Kollmann (2013), *Estimating the state vector of linearized DSGE models without the Kalman filter*. Handout §2.2.

Inversion solves directly for the shock

$$Y_t = y_t = Hx_t = HFx_{t-1} + HG\varepsilon_t$$

$$\varepsilon_t = (HG)^{-1}\left(Y_t - HFx_{t-1}\right), \qquad x_t = Fx_{t-1} + G\varepsilon_t$$

- The forecast error $Y_t - HFx_{t-1}$ is the economic surprise, rescaled to shock units through $(HG)^{-1}$.[9]
- The recovered shock then generates the next state: a conditional recursion, not a covariance update.
- This displayed inverse is the linear case. Nonlinear inversion solves the corresponding nonlinear observation relation. Both require admissible initialization ($x_0 = 0$ here).

[9] Inversion filter of Kollmann (2013), *Estimating the state vector of linearized DSGE models without the Kalman filter*. Handout §2.2.

Inversion solves directly for the shock

$$Y_t = y_t = Hx_t = HFx_{t-1} + HG\varepsilon_t$$

$$\varepsilon_t = (HG)^{-1}\left(Y_t - HFx_{t-1}\right), \qquad x_t = Fx_{t-1} + G\varepsilon_t$$

- The forecast error $Y_t - HFx_{t-1}$ is the economic surprise, rescaled to shock units through $(HG)^{-1}$.[9]
- The recovered shock then generates the next state: a conditional recursion, not a covariance update.
- This displayed inverse is the linear case. Nonlinear inversion solves the corresponding nonlinear observation relation. Both require admissible initialization ($x_0 = 0$ here).

[9] Inversion filter of Kollmann (2013), *Estimating the state vector of linearized DSGE models without the Kalman filter*. Handout §2.2.

When does direct inversion exist?

- No measurement error, so $Y_t = Hx_t$.
- A known initial state, usually $x_0 = 0$ in deviations.
- As many observables as shocks, and HG square nonsingular.
- Equal dimensions alone do not ensure independent shock transmission.

$$\det(HG) \neq 0$$

With more shocks than observables the direct inverse does not exist, but the Kalman filter remains applicable (handout §2.4).

When does direct inversion exist?

- No measurement error, so $Y_t = Hx_t$.
- A known initial state, usually $x_0 = 0$ in deviations.
- As many observables as shocks, and HG square nonsingular.
- Equal dimensions alone do not ensure independent shock transmission.

$$\det(HG) \neq 0$$

With more shocks than observables the direct inverse does not exist, but the Kalman filter remains applicable (handout §2.4).

When does direct inversion exist?

- No measurement error, so $Y_t = Hx_t$.
- A known initial state, usually $x_0 = 0$ in deviations.
- As many observables as shocks, and HG square nonsingular.
- Equal dimensions alone do not ensure independent shock transmission.

$$\det(HG) \neq 0$$

With more shocks than observables the direct inverse does not exist, but the Kalman filter remains applicable (handout §2.4).

When does direct inversion exist?

- No measurement error, so $Y_t = Hx_t$.
- A known initial state, usually $x_0 = 0$ in deviations.
- As many observables as shocks, and HG square nonsingular.
- Equal dimensions alone do not ensure independent shock transmission.

$$\det(HG) \neq 0$$

With more shocks than observables the direct inverse does not exist, but the Kalman filter remains applicable (handout §2.4).

When does direct inversion exist?

- No measurement error, so $Y_t = Hx_t$.
- A known initial state, usually $x_0 = 0$ in deviations.
- As many observables as shocks, and HG square nonsingular.
- Equal dimensions alone do not ensure independent shock transmission.

$$\det(HG) \neq 0$$

With more shocks than observables the direct inverse does not exist, but the Kalman filter remains applicable (handout §2.4).

When does direct inversion exist?

- No measurement error, so $Y_t = Hx_t$.
- A known initial state, usually $x_0 = 0$ in deviations.
- As many observables as shocks, and HG square nonsingular.
- Equal dimensions alone do not ensure independent shock transmission.

$$\det(HG) \neq 0$$

With more shocks than observables the direct inverse does not exist, but the Kalman filter remains applicable (handout §2.4).

Recover the artificial shocks

```
% After running invertRBC.m
max(abs(e_hat(:)-e_(:)))                % shock recovery
max(abs(y_obs(:)-reshape(H*x_hat,[],1))) % data replication
rank(H*G)                               % invertibility check
```

- The inversion loop is two lines: `e_hat(:,t) = (H*G)(y_obs(:,t-1)-H*F*x_hat(:,t-1))` then `x_hat(:,t) = F*x_hat(:,t-1)+G*e_hat(:,t)`.
- Explain why both reconstruction errors are near numerical precision.
- Would an exact fit still prove that the chosen economic model is correct?

Recover the artificial shocks

```
% After running invertRBC.m
max(abs(e_hat(:)-e_(:)))                % shock recovery
max(abs(y_obs(:)-reshape(H*x_hat,[],1))) % data replication
rank(H*G)                               % invertibility check
```

- The inversion loop is two lines: `e_hat(:,t) = (H*G)(y_obs(:,t-1)-H*F*x_hat(:,t-1))` then `x_hat(:,t) = F*x_hat(:,t-1)+G*e_hat(:,t)`.
- Explain why both reconstruction errors are near numerical precision.
- Would an exact fit still prove that the chosen economic model is correct?

Recover the artificial shocks

```
% After running invertRBC.m
max(abs(e_hat(:)-e_(:)))                % shock recovery
max(abs(y_obs(:)-reshape(H*x_hat,[],1))) % data replication
rank(H*G)                               % invertibility check
```

- The inversion loop is two lines: `e_hat(:,t) = (H*G)(y_obs(:,t-1)-H*F*x_hat(:,t-1))` then `x_hat(:,t) = F*x_hat(:,t-1)+G*e_hat(:,t)`.
- Explain why both reconstruction errors are near numerical precision.
- Would an exact fit still prove that the chosen economic model is correct?

Exact fit is built into inversion

$$Hx_t = HFx_{t-1} + HG\left(HG\right)^{-1}\left(Y_t - HFx_{t-1}\right) = Y_t$$

- The identity holds for any admissible parameter vector with invertible impact HG.
- The likelihood distinguishes parameters through shock densities and the observation Jacobian.
- Perfect reconstruction is a useful code check, not an independent economic test.

Exact fit is built into inversion

$$Hx_t = HFx_{t-1} + HG\left(HG\right)^{-1}\left(Y_t - HFx_{t-1}\right) = Y_t$$

- The identity holds for any admissible parameter vector with invertible impact HG.
- The likelihood distinguishes parameters through shock densities and the observation Jacobian.
- Perfect reconstruction is a useful code check, not an independent economic test.

Exact fit is built into inversion

$$Hx_t = HFx_{t-1} + HG\,(HG)^{-1}\,(Y_t - HFx_{t-1}) = Y_t$$

- The identity holds for any admissible parameter vector with invertible impact HG.
- The likelihood distinguishes parameters through shock densities and the observation Jacobian.
- Perfect reconstruction is a useful code check, not an independent economic test.

Choose the filter for the problem

Filter	Main condition and object
Kalman	Linear Gaussian system, state means and covariances
Inversion	Known initial state, no noise, invertible impact
Particle	Numerical alternative for nonlinear or non-Gaussian models

- The particle filter extends likelihood inference beyond the linear Gaussian case.[10]
- More shocks than observables rule out direct inversion, not Kalman filtering.

[10] Fernández-Villaverde and Rubio-Ramírez (2007), *Estimating macroeconomic models: A likelihood approach*, handout §2.3, at a substantially higher computational cost.

Choose the filter for the problem

Filter	Main condition and object
Kalman	Linear Gaussian system, state means and covariances
Inversion	Known initial state, no noise, invertible impact
Particle	Numerical alternative for nonlinear or non-Gaussian models

- The particle filter extends likelihood inference beyond the linear Gaussian case.[10]
- More shocks than observables rule out direct inversion, not Kalman filtering.

[10] Fernández-Villaverde and Rubio-Ramírez (2007), *Estimating macroeconomic models: A likelihood approach*, handout §2.3, at a substantially higher computational cost.

Plan

1. Introduction
2. Filters
3. Maximum likelihood
4. Estimation with Dynare

Prediction errors are available. Their conditional density now ranks candidate parameters.

Likelihood: a function of the parameters

Definition: likelihood function

For independent observations $x_1, \ldots, x_T$ in the Gaussian toy model and parameters θ,

$$L(\theta) = \prod_{t=1}^{T} f\left(x_t \mid \theta\right), \qquad \hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \; L(\theta)$$

- Toy forecasting model: $x_t = \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(\mu, \sigma^2)$, so $S_t = \varepsilon_t - \mu \sim \mathcal{N}(0, \sigma^2)$.
- For continuous data, maximize a density: the probability of any exact realization is zero.[11]
- θ becomes the independent variable: the data are fixed while μ and σ move.

[11] Handout §3.1. The notation $P(x_1, \ldots, x_T \mid \theta)$ must therefore be read as a density $f(x_1, \ldots, x_T \mid \theta)$.

Likelihood: a function of the parameters

Definition: likelihood function

For independent observations $x_1, \ldots, x_T$ in the Gaussian toy model and parameters θ,

$$L(\theta) = \prod_{t=1}^{T} f\left(x_t \mid \theta\right), \qquad \hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \; L(\theta)$$

- Toy forecasting model: $x_t = \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(\mu, \sigma^2)$, so $S_t = \varepsilon_t - \mu \sim \mathcal{N}(0, \sigma^2)$.
- For continuous data, maximize a density: the probability of any exact realization is zero.[11]
- θ becomes the independent variable: the data are fixed while μ and σ move.

[11] Handout §3.1. The notation $P(x_1, \ldots, x_T \mid \theta)$ must therefore be read as a density $f(x_1, \ldots, x_T \mid \theta)$.

Likelihood: a function of the parameters

Definition: likelihood function

For independent observations $x_1, \ldots, x_T$ in the Gaussian toy model and parameters θ,

$$L(\theta) = \prod_{t=1}^{T} f\left(x_t \mid \theta\right), \qquad \hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \; L(\theta)$$

- Toy forecasting model: $x_t = \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(\mu, \sigma^2)$, so $S_t = \varepsilon_t - \mu \sim \mathcal{N}(0, \sigma^2)$.
- For continuous data, maximize a density: the probability of any exact realization is zero.[11]
- θ becomes the independent variable: the data are fixed while μ and σ move.

[11] Handout §3.1. The notation $P(x_1, \ldots, x_T \mid \theta)$ must therefore be read as a density $f(x_1, \ldots, x_T \mid \theta)$.

Which candidate density fits the sample?

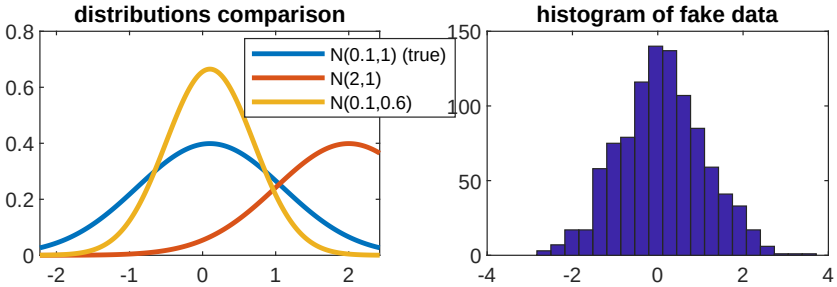
$$f(\varepsilon_t \mid \theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(\varepsilon_t - \mu)^2}{2\sigma^2}\right]$$

Candidate	μ	σ	Verdict
True DGP	0.1	1	reference
θ'	2	1	shifted
θ''	0.1	0.6	too narrow

- $T = 1000$ draws from $\mathcal{N}(0.1, 1)$, three Gaussian curves compared to the histogram.[12]
- The likelihood ranks the curves: $L(Y, \theta) > L(Y, \theta')$ and $L(Y, \theta) > L(Y, \theta'')$.
- `LK.m` keeps the underflowing products and also sums analytic log densities to rank these candidates reliably.

[12] `LK.m` illustration, handout §3.1.

Compare densities with the actual sample



The left panel changes the candidate mean or standard deviation. The right panel fixes the artificial sample. The likelihood ranks the candidate densities on these same observations.[13]

[13] Handout MATLAB box 1, `LK.m`. Original illustrative simulation, not a new sample.

From densities to negative log-likelihood

$$\log \prod_{t=1}^{T} f\left(x_t \mid \theta\right) = \sum_{t=1}^{T} \log f(x_t \mid \theta) = \sum_{t=1}^{T} \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(\varepsilon_t - \mu)^2}{2\sigma^2} \right]$$

$$= -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^{T} (\varepsilon_t - \mu)^2$$

- Logs turn products into sums, which are easier to code and to differentiate.[14]
- Software minimizes more readily than it maximizes:

$$\hat{\theta}_{\text{MLE}} = \arg\min_{\{\theta\}} \left\{ -\log f\left(x_1, x_2, \ldots, x_T \mid \theta\right) \right\}$$

- The maximizer is unchanged by the sign flip.

[14] Handout §3.1: the two standard tricks before any numerical optimization.

From densities to negative log-likelihood

$$\log \prod_{t=1}^{T} f\left(x_t \mid \theta\right) = \sum_{t=1}^{T} \log f(x_t \mid \theta) = \sum_{t=1}^{T} \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(\varepsilon_t - \mu)^2}{2\sigma^2} \right]$$

$$= -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^{T} (\varepsilon_t - \mu)^2$$

- Logs turn products into sums, which are easier to code and to differentiate.[14]
- Software minimizes more readily than it maximizes:

$$\hat{\theta}_{\text{MLE}} = \arg\min_{\{\theta\}} \left\{ -\log f\left(x_1, x_2, \ldots, x_T \mid \theta\right) \right\}$$

- The maximizer is unchanged by the sign flip.

[14] Handout §3.1: the two standard tricks before any numerical optimization.

From densities to negative log-likelihood

$$\log \prod_{t=1}^{T} f\left(x_t \mid \theta\right) = \sum_{t=1}^{T} \log f(x_t \mid \theta) = \sum_{t=1}^{T} \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(\varepsilon_t - \mu)^2}{2\sigma^2} \right]$$

$$= -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^{T} (\varepsilon_t - \mu)^2$$

- Logs turn products into sums, which are easier to code and to differentiate.[14]
- Software minimizes more readily than it maximizes:

$$\hat{\theta}_{\text{MLE}} = \arg\min_{\{\theta\}} \left\{ -\log f\left(x_1, x_2, \ldots, x_T \mid \theta\right) \right\}$$

- The maximizer is unchanged by the sign flip.

[14] Handout §3.1: the two standard tricks before any numerical optimization.

The Gaussian MLE benchmark

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_t (x_t - \mu) = 0 \quad \Longrightarrow \quad \hat{\mu} = \bar{x}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{T}{2\sigma^2} + \frac{\sum_t (x_t - \hat{\mu})^2}{2(\sigma^2)^2} = 0 \quad \Longrightarrow \quad \widehat{\sigma^2} = \frac{1}{T}\sum_{t=1}^{T}(x_t - \bar{x})^2$$

- The MLE divides by T, not $T-1$: the estimator of σ^2 differs from the sample-variance convention.[15]
- Use these closed forms to check the numerical optimizer of `MLE.m`.
- Same principle with n observables: differentiate the log-likelihood of the filter outputs.

[15] Derivation added as a benchmark for the handout's numerical exercise, consistent with handout §3.1.

The Gaussian MLE benchmark

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_t (x_t - \mu) = 0 \quad \Longrightarrow \quad \hat{\mu} = \bar{x}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{T}{2\sigma^2} + \frac{\sum_t (x_t - \hat{\mu})^2}{2(\sigma^2)^2} = 0 \quad \Longrightarrow \quad \widehat{\sigma^2} = \frac{1}{T}\sum_{t=1}^{T}(x_t - \bar{x})^2$$

- The MLE divides by T, not $T-1$: the estimator of σ^2 differs from the sample-variance convention.[15]
- Use these closed forms to check the numerical optimizer of `MLE.m`.
- Same principle with n observables: differentiate the log-likelihood of the filter outputs.

[15] Derivation added as a benchmark for the handout's numerical exercise, consistent with handout §3.1.

The Gaussian MLE benchmark

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_t (x_t - \mu) = 0 \quad \Longrightarrow \quad \hat{\mu} = \bar{x}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{T}{2\sigma^2} + \frac{\sum_t (x_t - \hat{\mu})^2}{2(\sigma^2)^2} = 0 \quad \Longrightarrow \quad \widehat{\sigma^2} = \frac{1}{T}\sum_{t=1}^{T}(x_t - \bar{x})^2$$

- The MLE divides by T, not $T-1$: the estimator of σ^2 differs from the sample-variance convention.[15]
- Use these closed forms to check the numerical optimizer of `MLE.m`.
- Same principle with n observables: differentiate the log-likelihood of the filter outputs.

[15] Derivation added as a benchmark for the handout's numerical exercise, consistent with handout §3.1.

Implement the Gaussian MLE

```
mu = 0.1; sig = 1; T = 5000;
y_obs = mu + sig*randn(T,1);          % true DGP N(0.1,1)
llk = @(x) -( -T/2*log(2*pi*x(2)^2) ...
    - 1/(2*x(2)^2)*(y_obs-x(1))'*(y_obs-x(1)) );
theta_MLE = fmincon(llk,[0 5],[],[],[],[],[-Inf 1e-8]);
```

- `llk` returns the *negative* log-likelihood, so a minimizer solves the original maximization.
- The lower bound 10^{-8} keeps σ positive.
- The script also profiles the objective in μ and in σ: compare the optimizer with $\hat{\mu} = \bar{x}$ and $\widehat{\sigma^2} = T^{-1}\sum(x_t - \bar{x})^2$.

Implement the Gaussian MLE

```
mu = 0.1; sig = 1; T = 5000;
y_obs = mu + sig*randn(T,1);          % true DGP N(0.1,1)
llk = @(x) -( -T/2*log(2*pi*x(2)^2) ...
    - 1/(2*x(2)^2)*(y_obs-x(1))'*(y_obs-x(1)) );
theta_MLE = fmincon(llk,[0 5],[],[],[],[],[-Inf 1e-8]);
```

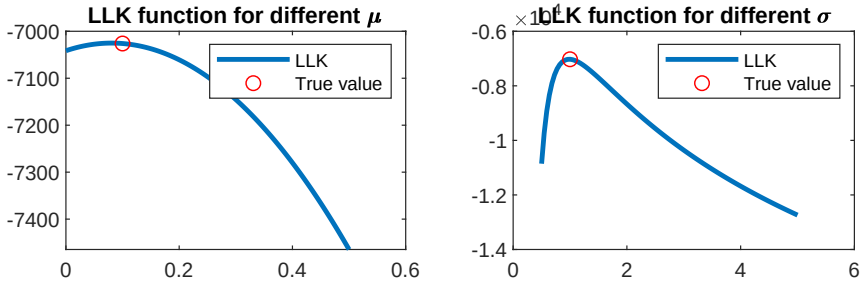
- `llk` returns the *negative* log-likelihood, so a minimizer solves the original maximization.
- The lower bound 10^{-8} keeps σ positive.
- The script also profiles the objective in μ and in σ: compare the optimizer with $\hat{\mu} = \bar{x}$ and $\widehat{\sigma^2} = T^{-1}\sum(x_t - \bar{x})^2$.

Implement the Gaussian MLE

```
mu = 0.1; sig = 1; T = 5000;
y_obs = mu + sig*randn(T,1);          % true DGP N(0.1,1)
llk = @(x) -( -T/2*log(2*pi*x(2)^2) ...
    - 1/(2*x(2)^2)*(y_obs-x(1))'*(y_obs-x(1)) );
theta_MLE = fmincon(llk,[0 5],[],[],[],[],[-Inf 1e-8]);
```

- `llk` returns the *negative* log-likelihood, so a minimizer solves the original maximization.
- The lower bound 10^{-8} keeps σ positive.
- The script also profiles the objective in μ and in σ: compare the optimizer with $\hat{\mu} = \bar{x}$ and $\widehat{\sigma^2} = T^{-1}\sum(x_t - \bar{x})^2$.

Inspect the likelihood profiles



The profiles vary the mean and standard deviation separately. Their maxima should agree with the analytic benchmark, up to sampling variation and the held-fixed parameter.[16]

[16] Handout MATLAB box 2, `MLE.m`. Original illustration. The graph shows log-likelihood, whereas the optimizer minimizes its negative.

The Gaussian DSGE likelihood

Definition: Gaussian log-likelihood from the Kalman filter

With n observed variables and $S_t \sim \mathcal{N}(0, \Omega_t)$:

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|\Omega_t|\right) - \frac{1}{2}\sum_{t=1}^{T} S_t'\Omega_t^{-1}S_t$$

- n: number of observed series, θ: estimated parameters (H, F, G, R, Q are functions of θ).
- The quadratic term penalizes large standardized forecast errors.
- The determinant term penalizes excessive forecast uncertainty: a model that expects nothing cannot hide surprises.
- Each candidate θ requires re-solving and re-filtering the full sample.

The Gaussian DSGE likelihood

Definition: Gaussian log-likelihood from the Kalman filter

With n observed variables and $S_t \sim \mathcal{N}(0, \Omega_t)$:

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|\Omega_t|\right) - \frac{1}{2}\sum_{t=1}^{T} S_t'\Omega_t^{-1}S_t$$

- n: number of observed series, θ: estimated parameters (H, F, G, R, Q are functions of θ).
- The quadratic term penalizes large standardized forecast errors.
- The determinant term penalizes excessive forecast uncertainty: a model that expects nothing cannot hide surprises.
- Each candidate θ requires re-solving and re-filtering the full sample.

The Gaussian DSGE likelihood

Definition: Gaussian log-likelihood from the Kalman filter

With n observed variables and $S_t \sim \mathcal{N}(0, \Omega_t)$:

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|\Omega_t|\right) - \frac{1}{2}\sum_{t=1}^{T} S_t'\Omega_t^{-1}S_t$$

- n: number of observed series, θ: estimated parameters (H, F, G, R, Q are functions of θ).
- The quadratic term penalizes large standardized forecast errors.
- The determinant term penalizes excessive forecast uncertainty: a model that expects nothing cannot hide surprises.
- Each candidate θ requires re-solving and re-filtering the full sample.

The Gaussian DSGE likelihood

Definition: Gaussian log-likelihood from the Kalman filter

With n observed variables and $S_t \sim \mathcal{N}(0, \Omega_t)$:

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|\Omega_t|\right) - \frac{1}{2}\sum_{t=1}^{T} S_t'\Omega_t^{-1}S_t$$

- n: number of observed series, θ: estimated parameters (H, F, G, R, Q are functions of θ).
- The quadratic term penalizes large standardized forecast errors.
- The determinant term penalizes excessive forecast uncertainty: a model that expects nothing cannot hide surprises.
- Each candidate θ requires re-solving and re-filtering the full sample.

Inversion likelihood and the Jacobian

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|Q|\right) - \frac{1}{2}\sum_{t=1}^{T}\varepsilon_t' Q^{-1}\varepsilon_t - T\log|\det(HG)|$$

- Recovered shocks satisfy $\varepsilon_t \sim \mathcal{N}(0, Q)$ only under the inversion conditions.[17]
- Mapping shocks to observations rescales the density by $|\det(HG)|^{-1}$ at each date: the Jacobian term.
- If only Q is estimated with fixed F, G, H, the Jacobian is constant.
- If estimated parameters enter G, omitting the Jacobian can move the estimate.

[17] Known initial state, no measurement error, invertible HG (handout §3.2).

Inversion likelihood and the Jacobian

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|Q|\right) - \frac{1}{2}\sum_{t=1}^{T}\varepsilon_t' Q^{-1}\varepsilon_t - T\log|\det(HG)|$$

- Recovered shocks satisfy $\varepsilon_t \sim \mathcal{N}(0, Q)$ only under the inversion conditions.[17]
- Mapping shocks to observations rescales the density by $|\det(HG)|^{-1}$ at each date: the Jacobian term.
- If only Q is estimated with fixed F, G, H, the Jacobian is constant.
- If estimated parameters enter G, omitting the Jacobian can move the estimate.

[17] Known initial state, no measurement error, invertible HG (handout §3.2).

Inversion likelihood and the Jacobian

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|Q|\right) - \frac{1}{2}\sum_{t=1}^{T}\varepsilon_t' Q^{-1}\varepsilon_t - T\log|\det(HG)|$$

- Recovered shocks satisfy $\varepsilon_t \sim \mathcal{N}(0, Q)$ only under the inversion conditions.[17]
- Mapping shocks to observations rescales the density by $|\det(HG)|^{-1}$ at each date: the Jacobian term.
- If only Q is estimated with fixed F, G, H, the Jacobian is constant.
- If estimated parameters enter G, omitting the Jacobian can move the estimate.

[17] Known initial state, no measurement error, invertible HG (handout §3.2).

Inversion likelihood and the Jacobian

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\left(|Q|\right) - \frac{1}{2}\sum_{t=1}^{T}\varepsilon_t' Q^{-1}\varepsilon_t - T\log|\det(HG)|$$

- Recovered shocks satisfy $\varepsilon_t \sim \mathcal{N}(0, Q)$ only under the inversion conditions.[17]
- Mapping shocks to observations rescales the density by $|\det(HG)|^{-1}$ at each date: the Jacobian term.
- If only Q is estimated with fixed F, G, H, the Jacobian is constant.
- If estimated parameters enter G, omitting the Jacobian can move the estimate.

[17] Known initial state, no measurement error, invertible HG (handout §3.2).

Estimate the artificial RBC economy

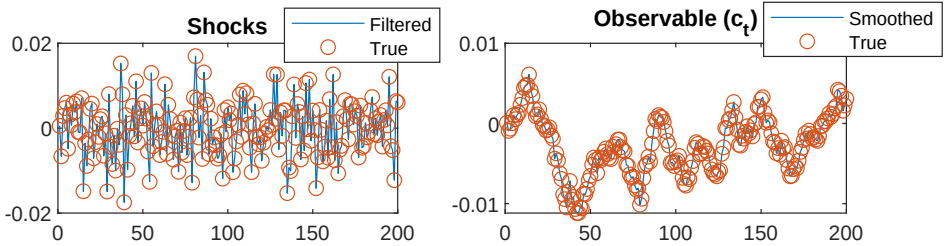
- $T = 200$ artificial observations, one technology shock, consumption observed: one shock must back one observable.
- Inversion retrieves the generating shock sequence: feeding the shocks back reproduces consumption exactly.[18]

$$\hat{Q} = \frac{1}{T}\sum_{t} \hat{\varepsilon}_t^2, \qquad \hat{\sigma}_\varepsilon = \sqrt{\hat{Q}}$$

- Historical result: $\hat{\sigma}_\varepsilon = 0.0067$ against true $\sigma_\varepsilon = 0.007$.[19]
- The Kalman pass gives a similar estimate: the scripts optimize $\log Q$ to enforce positivity. Their initial-state assumptions differ.

[18] Handout §3.3, `invertRBC.m` illustration.
[19] Handout §3.3: historical illustration, not a new run.

Shocks and consumption reconstructed



Known initial conditions and an invertible impact recover the generating shocks. Feeding them back reproduces the observed consumption series.[20]

[20] Handout MATLAB box 3, `invertRBC.m`, 200 observations. Historical simulation, not a new execution.

Plan

1. Introduction
2. Filters
3. Maximum likelihood
4. Estimation with Dynare

The criterion is defined. Align it with transformed data and examine identification.

Why real GDP must be transformed

- Macroeconomic levels may contain stochastic or deterministic trends. Removing a linear trend need not make them stationary.[21]
- Two forces make GDP grow steadily: technological change and population growth.[22]
- Remedies: divide by the working population, or rewrite the model in effective labour units, then remove the technology trend.
- The solved model lives in stationary deviations: the data must be brought to the same space before estimation.

[21] Nelson and Plosser (1982), *Trends and random walks in macroeconomic time series*, handout §4.1.
[22] Solow (1956), *A contribution to the theory of economic growth.*

Why real GDP must be transformed

- Macroeconomic levels may contain stochastic or deterministic trends. Removing a linear trend need not make them stationary.[21]
- Two forces make GDP grow steadily: technological change and population growth.[22]
- Remedies: divide by the working population, or rewrite the model in effective labour units, then remove the technology trend.
- The solved model lives in stationary deviations: the data must be brought to the same space before estimation.

[21] Nelson and Plosser (1982), *Trends and random walks in macroeconomic time series*, handout §4.1.
[22] Solow (1956), *A contribution to the theory of economic growth.*

Why real GDP must be transformed

- Macroeconomic levels may contain stochastic or deterministic trends. Removing a linear trend need not make them stationary.[21]
- Two forces make GDP grow steadily: technological change and population growth.[22]
- Remedies: divide by the working population, or rewrite the model in effective labour units, then remove the technology trend.
- The solved model lives in stationary deviations: the data must be brought to the same space before estimation.

[21] Nelson and Plosser (1982), *Trends and random walks in macroeconomic time series*, handout §4.1.
[22] Solow (1956), *A contribution to the theory of economic growth.*

Why real GDP must be transformed

- Macroeconomic levels may contain stochastic or deterministic trends. Removing a linear trend need not make them stationary.[21]
- Two forces make GDP grow steadily: technological change and population growth.[22]
- Remedies: divide by the working population, or rewrite the model in effective labour units, then remove the technology trend.
- The solved model lives in stationary deviations: the data must be brought to the same space before estimation.

[21] Nelson and Plosser (1982), *Trends and random walks in macroeconomic time series*, handout §4.1.
[22] Solow (1956), *A contribution to the theory of economic growth.*

Identify and validate the data

1. Verify provider, dataset, dimensions and series code on DBnomics.
2. Use the standard DBnomics MATLAB client: `[my_mat,table,dates_nb] = call_dbnomics(...)` in `mydb.m`.
3. Convert column 1 of `my_mat` from datenums to datetime. The third output is numeric, not datetime. Check frequency, units, duplicates and missing periods.
4. Record identifier, retrieval date, sample and raw data for reproducibility.

Historical series: `OECD/QNA/USA.B1_GS1.LNBQRSA.Q`

Before Dynare: run `mydb.m` so that `mydata.mat` exists with the observed variable y.

Identify and validate the data

1. Verify provider, dataset, dimensions and series code on DBnomics.
2. Use the standard DBnomics MATLAB client: `[my_mat,table,dates_nb] = call_dbnomics(...)` in `mydb.m`.
3. Convert column 1 of `my_mat` from datenums to datetime. The third output is numeric, not datetime. Check frequency, units, duplicates and missing periods.
4. Record identifier, retrieval date, sample and raw data for reproducibility.

Historical series: `OECD/QNA/USA.B1_GS1.LNBQRSA.Q`

Before Dynare: run `mydb.m` so that `mydata.mat` exists with the observed variable y.

Identify and validate the data

1. Verify provider, dataset, dimensions and series code on DBnomics.
2. Use the standard DBnomics MATLAB client: `[my_mat,table,dates_nb] = call_dbnomics(...)` in `mydb.m`.
3. Convert column 1 of `my_mat` from datenums to datetime. The third output is numeric, not datetime. Check frequency, units, duplicates and missing periods.
4. Record identifier, retrieval date, sample and raw data for reproducibility.

Historical series: `OECD/QNA/USA.B1_GS1.LNBQRSA.Q`

Before Dynare: run `mydb.m` so that `mydata.mat` exists with the observed variable y.

Identify and validate the data

1. Verify provider, dataset, dimensions and series code on DBnomics.
2. Use the standard DBnomics MATLAB client: `[my_mat,table,dates_nb] = call_dbnomics(...)` in `mydb.m`.
3. Convert column 1 of `my_mat` from datenums to datetime. The third output is numeric, not datetime. Check frequency, units, duplicates and missing periods.
4. Record identifier, retrieval date, sample and raw data for reproducibility.

Historical series: `OECD/QNA/USA.B1_GS1.LNBQRSA.Q`

Before Dynare: run `mydb.m` so that `mydata.mat` exists with the observed variable y.

Identify and validate the data

1. Verify provider, dataset, dimensions and series code on DBnomics.
2. Use the standard DBnomics MATLAB client: `[my_mat,table,dates_nb] = call_dbnomics(...)` in `mydb.m`.
3. Convert column 1 of `my_mat` from datenums to datetime. The third output is numeric, not datetime. Check frequency, units, duplicates and missing periods.
4. Record identifier, retrieval date, sample and raw data for reproducibility.

Historical series: `OECD/QNA/USA.B1_GS1.LNBQRSA.Q`

Before Dynare: run `mydb.m` so that `mydata.mat` exists with the observed variable y.

Identify and validate the data

1. Verify provider, dataset, dimensions and series code on DBnomics.
2. Use the standard DBnomics MATLAB client: `[my_mat,table,dates_nb] = call_dbnomics(...)` in `mydb.m`.
3. Convert column 1 of `my_mat` from datenums to datetime. The third output is numeric, not datetime. Check frequency, units, duplicates and missing periods.
4. Record identifier, retrieval date, sample and raw data for reproducibility.

Historical series: `OECD/QNA/USA.B1_GS1.LNBQRSA.Q`

Before Dynare: run `mydb.m` so that `mydata.mat` exists with the observed variable y.

Remove a log-linear trend in three steps

$$\hat{x}_t \simeq \log\left(\frac{X_t}{\bar{X}_0 \gamma^t}\right) \qquad \text{so} \qquad \hat{X}_t \simeq \log X_t - \log \gamma^t, \quad \hat{x}_t = \hat{X}_t - \mu(\hat{X}_t)$$

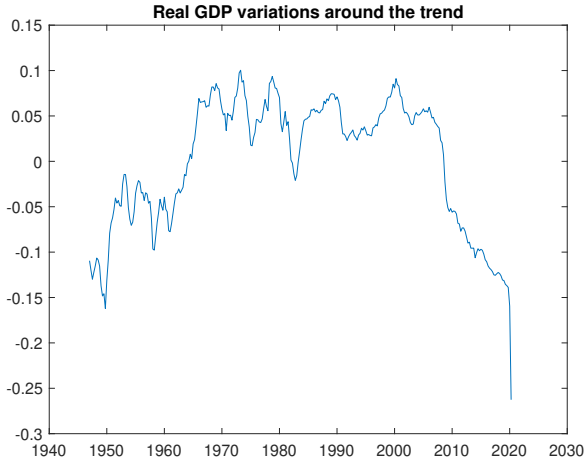
```
ln_y = log(gross_y);                % (a) log
X = [ones(length(ln_y),1) ...
     (1:length(ln_y))'];
OLS = X\ln_y;                       % (b)
    trend
trend_y = X*OLS;
gamma = exp(OLS(2));
```

```
ln_gap = ln_y-trend_y;
y = ln_gap-mean(ln_gap);            % (c)
    mean
```

- Regression $y_t = b_0 + b_1 t + \epsilon_t$ with $b_1 = \log \gamma$.
- $\gamma = 1.005$ means roughly 0.5% quarterly growth: sample-dependent.[23]

[23] Handout §4.1. With an intercept, the residual mean is already zero, demeaning is a safeguard.

US output around its fitted trend



- The gap is relative to the imposed log-linear trend, not an observation of potential output.
- Can productivity shocks alone explain this gap?[24]

[24] Handout figure 1. Historical illustration, not a new estimation.

Dynare: the model and its output observable

```
model(linear);
  [name='Budget constraint']
  k = Kss^(alpha-1)*a
      +(alpha*Kss^(alpha-1)+1-delta)*k(-1)-Css/Kss*c;
  [name='Euler equation']
  c = c(+1)-beta*alpha*Kss^(alpha-1)*(a(+1)+(alpha-1)*k);
  [name='Productivity shock']
  a = rho_A*a(-1)+e_a;
  [name='Production']
  y = a+alpha*k(-1);
end;
```

- The budget constraint and the Euler equation determine capital and consumption jointly.
- Here `Kss` and `Css` are fixed parameters: they move only in the deep-parameters exercise.
- One innovation `e_a`, one observable `y`.

Dynare: the model and its output observable

```
model(linear);
  [name='Budget constraint']
  k = Kss^(alpha-1)*a
      +(alpha*Kss^(alpha-1)+1-delta)*k(-1)-Css/Kss*c;
  [name='Euler equation']
  c = c(+1)-beta*alpha*Kss^(alpha-1)*(a(+1)+(alpha-1)*k);
  [name='Productivity shock']
  a = rho_A*a(-1)+e_a;
  [name='Production']
  y = a+alpha*k(-1);
end;
```

- The budget constraint and the Euler equation determine capital and consumption jointly.
- Here `Kss` and `Css` are fixed parameters: they move only in the deep-parameters exercise.
- One innovation `e_a`, one observable `y`.

Dynare: the model and its output observable

```
model(linear);
  [name='Budget constraint']
  k = Kss^(alpha-1)*a
      +(alpha*Kss^(alpha-1)+1-delta)*k(-1)-Css/Kss*c;
  [name='Euler equation']
  c = c(+1)-beta*alpha*Kss^(alpha-1)*(a(+1)+(alpha-1)*k);
  [name='Productivity shock']
  a = rho_A*a(-1)+e_a;
  [name='Production']
  y = a+alpha*k(-1);
end;
```

- The budget constraint and the Euler equation determine capital and consumption jointly.
- Here `Kss` and `Css` are fixed parameters: they move only in the deep-parameters exercise.
- One innovation `e_a`, one observable `y`.

Declare observables and parameters

```
estimated_params;
  stderr e_a, .007, , inf;     % initial .007, bounds
  rho_A, .95, 0, .999;         % initial .95, bounds
end;
varobs y;                        % observed variables
estimation(datafile=mydata,smoother) y c k a;
```

- Syntax: `parameter, initial value [, lower bound, upper bound]`. `stderr` targets a shock standard deviation.[25]
- `varobs` must match variables present in the data file, otherwise the estimation does not start.
- `smoother` asks for smoothed states and shocks on top of the likelihood.

[25] Handout §4.2. Initial values and bounds are *not* priors: this is maximum likelihood.

Declare observables and parameters

```
estimated_params;
  stderr e_a, .007, , inf;     % initial .007, bounds
  rho_A, .95, 0, .999;         % initial .95, bounds
end;
varobs y;                        % observed variables
estimation(datafile=mydata,smoother) y c k a;
```

- Syntax: `parameter, initial value [, lower bound, upper bound]`. `stderr` targets a shock standard deviation.[25]
- `varobs` must match variables present in the data file, otherwise the estimation does not start.
- `smoother` asks for smoothed states and shocks on top of the likelihood.

[25] Handout §4.2. Initial values and bounds are *not* priors: this is maximum likelihood.

Declare observables and parameters

```
estimated_params;
  stderr e_a, .007, , inf;     % initial .007, bounds
  rho_A, .95, 0, .999;         % initial .95, bounds
end;
varobs y;                        % observed variables
estimation(datafile=mydata,smoother) y c k a;
```

- Syntax: `parameter, initial value [, lower bound, upper bound]`. `stderr` targets a shock standard deviation.[25]
- `varobs` must match variables present in the data file, otherwise the estimation does not start.
- `smoother` asks for smoothed states and shocks on top of the likelihood.

[25] Handout §4.2. Initial values and bounds are *not* priors: this is maximum likelihood.

Run the estimation in the right order

Start MATLAB in `session3/codes`, with Dynare and the standard DBnomics client on the path.

```
which call_dbnomics
mydb
load mydata.mat y T
assert(all(isfinite(y)) && numel(y)==numel(T))
dynare RBC_estim.mod
```

- Check quarterly continuity, units and the fitted trend before estimation.
- Inspect the optimizer exit, parameter bounds and smoothed shocks.
- Save data and output. Historical figures are illustrations, not promised results for the new sample.

Run the estimation in the right order

Start MATLAB in `session3/codes`, with Dynare and the standard DBnomics client on the path.

```
which call_dbnomics
mydb
load mydata.mat y T
assert(all(isfinite(y)) && numel(y)==numel(T))
dynare RBC_estim.mod
```

- Check quarterly continuity, units and the fitted trend before estimation.
- Inspect the optimizer exit, parameter bounds and smoothed shocks.
- Save data and output. Historical figures are illustrations, not promised results for the new sample.

Run the estimation in the right order

Start MATLAB in `session3/codes`, with Dynare and the standard DBnomics client on the path.

```
which call_dbnomics
mydb
load mydata.mat y T
assert(all(isfinite(y)) && numel(y)==numel(T))
dynare RBC_estim.mod
```

- Check quarterly continuity, units and the fitted trend before estimation.
- Inspect the optimizer exit, parameter bounds and smoothed shocks.
- Save data and output. Historical figures are illustrations, not promised results for the new sample.

Reading the Dynare output

Console excerpt (handout)

```
Improvement on iteration 10 = 3.1e-08
improvement < crit termination
Final value of minus log posterior:
-916.468634
```

Parameter	Estimate	s.e. (t)
ρ_A	0.8362	0.0413 (20.24)
σ_{e_a}	0.0106	0.0004 (24.26)

- The final value is *minus* the log-likelihood, so here $\log L \approx 916.47$. A density may exceed one.[26]
- $|t| > 1.96$ rejects a zero null in a two-sided 5% Wald test under regularity conditions.
- This rule is not an identification diagnostic, and variance parameters near a boundary need separate care.

[26] Handout §4.3 console excerpt: historical illustration, not a current-sample estimate.

Reading the Dynare output

Console excerpt (handout)

```
Improvement on iteration 10 = 3.1e-08
improvement < crit termination
Final value of minus log posterior:
-916.468634
```

Parameter	Estimate	s.e. (t)
ρ_A	0.8362	0.0413 (20.24)
σ_{e_a}	0.0106	0.0004 (24.26)

- The final value is *minus* the log-likelihood, so here $\log L \approx 916.47$. A density may exceed one.[26]
- $|t| > 1.96$ rejects a zero null in a two-sided 5% Wald test under regularity conditions.
- This rule is not an identification diagnostic, and variance parameters near a boundary need separate care.

[26] Handout §4.3 console excerpt: historical illustration, not a current-sample estimate.

Reading the Dynare output

Console excerpt (handout)

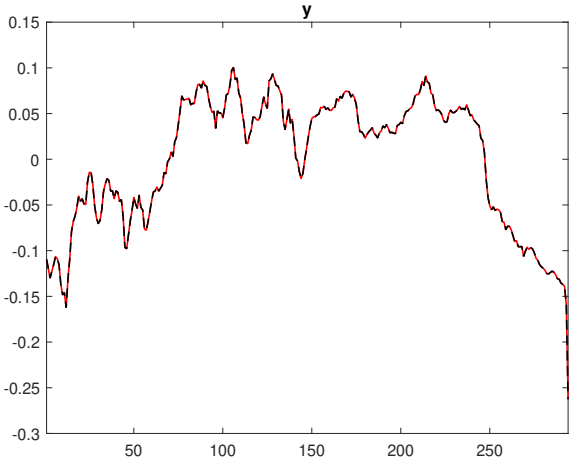
```
Improvement on iteration 10 = 3.1e-08
improvement < crit termination
Final value of minus log posterior:
-916.468634
```

Parameter	Estimate	s.e. (t)
ρ_A	0.8362	0.0413 (20.24)
σ_{e_a}	0.0106	0.0004 (24.26)

- The final value is *minus* the log-likelihood, so here $\log L \approx 916.47$. A density may exceed one.[26]
- $|t| > 1.96$ rejects a zero null in a two-sided 5% Wald test under regularity conditions.
- This rule is not an identification diagnostic, and variance parameters near a boundary need separate care.

[26] Handout §4.3 console excerpt: historical illustration, not a current-sample estimate.

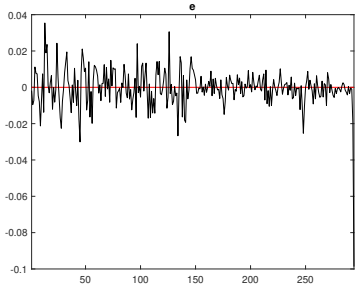
Observed and reconstructed output



- With one observable and one invertible shock, a near-exact fit can be mechanical.
- The likelihood still ranks the plausibility of the recovered shocks, not just the visual fit.[27]

[27] Handout figure 3. Observed series in black, reconstructed series in red. Historical illustration.

Smoothed shocks and stored results

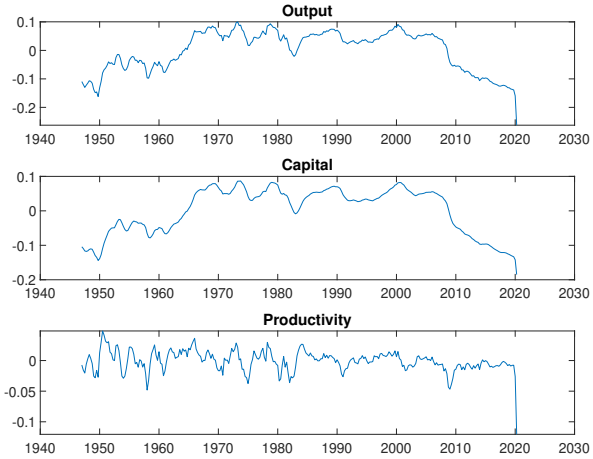


- `oo_.SmoothedShocks`: shocks conditional on the whole sample.[28]
- `oo_.SmoothedVariables`: corresponding endogenous paths.

Smoothing uses later observations to infer past states and shocks. It differs from a real-time forecast.

[28] Handout figure 2, historical illustration. The figure is not a new estimation result.

Latent paths are model-dependent objects



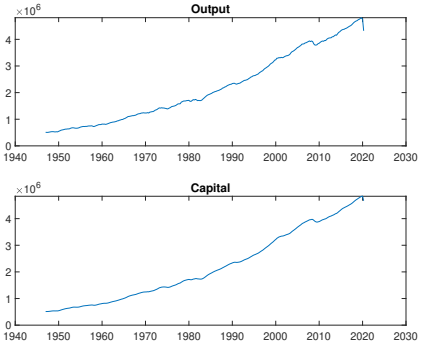
- After the crisis, persistent negative output gaps accompany lower inferred capital.
- Changing realized volatility is not proof of a structural break in innovation variance.
- These paths are model-dependent inferences, not separately observed data.[29]

[29] Handout figure 4. Historical model-generated output, capital and productivity.

Reverse the detrending with care

```
% REVERSING
figure;
subplot(2,1,1);
plot(T,exp(trend_y+...
     oo_.SmoothedVariables.y))
subplot(2,1,2);
plot(T,exp(trend_y-trend_y(1)+...
     oo_.SmoothedVariables.k))
```

- Add the trend back and invert the log for illustrative level paths.



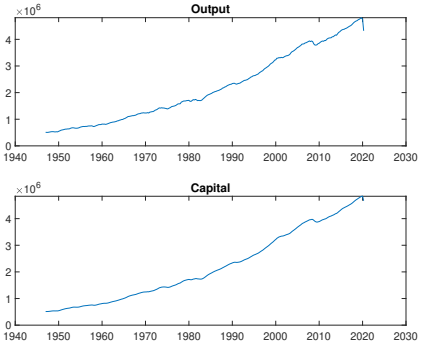
The capital panel is an index built with the GDP trend: recovering capital in economic units requires a capital-specific steady-state scale and trend.[30]

[30] Handout figure 5 caption. The productivity shock is stationary, so it has no trend to reverse.

Reverse the detrending with care

```
% REVERSING
figure;
subplot(2,1,1);
plot(T,exp(trend_y+...
     oo_.SmoothedVariables.y))
subplot(2,1,2);
plot(T,exp(trend_y-trend_y(1)+...
     oo_.SmoothedVariables.k))
```

- Add the trend back and invert the log for illustrative level paths.



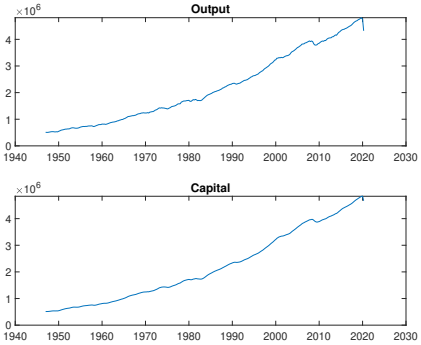
The capital panel is an index built with the GDP trend: recovering capital in economic units requires a capital-specific steady-state scale and trend.[30]

[30] Handout figure 5 caption. The productivity shock is stationary, so it has no trend to reverse.

Reverse the detrending with care

```
% REVERSING
figure;
subplot(2,1,1);
plot(T,exp(trend_y+...
     oo_.SmoothedVariables.y))
subplot(2,1,2);
plot(T,exp(trend_y-trend_y(1)+...
     oo_.SmoothedVariables.k))
```

- Add the trend back and invert the log for illustrative level paths.



The capital panel is an index built with the GDP trend: recovering capital in economic units requires a capital-specific steady-state scale and trend.[30]

[30] Handout figure 5 caption. The productivity shock is stationary, so it has no trend to reverse.

Read and work: moments and IRFs

Read handout 3, §4.3, Exercise 1, *Second moment statistics and IRFs of the estimated model*, then work through all four questions.

- Checkpoint: distinguish structural innovation s.d., dispersion of smoothed shocks and unconditional model volatility.
- Compare moments in identical units and with identical detrending.
- Bring one estimated-model IRF and explain why it is not a fitted historical path.

Work from `session3/codes`. Retain transformations, sample and estimates.

Read and work: moments and IRFs

Read handout 3, §4.3, Exercise 1, *Second moment statistics and IRFs of the estimated model*, then work through all four questions.

- Checkpoint: distinguish structural innovation s.d., dispersion of smoothed shocks and unconditional model volatility.
- Compare moments in identical units and with identical detrending.
- Bring one estimated-model IRF and explain why it is not a fitted historical path.

Work from `session3/codes`. Retain transformations, sample and estimates.

Read and work: moments and IRFs

Read handout 3, §4.3, Exercise 1, *Second moment statistics and IRFs of the estimated model*, then work through all four questions.

- Checkpoint: distinguish structural innovation s.d., dispersion of smoothed shocks and unconditional model volatility.
- Compare moments in identical units and with identical detrending.
- Bring one estimated-model IRF and explain why it is not a fitted historical path.

Work from `session3/codes`. Retain transformations, sample and estimates.

Public spending adds a demand channel

$$Y_t = C_t + I_t + G_t, \qquad G_t = \int_0^1 T_{jt}\,\mathrm{d}j, \qquad C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^{\alpha} + (1-\delta)\,K_{jt-1}$$

$$G_t = \bar{G}\,\varepsilon_t^G, \quad \bar{G} = g^y \bar{Y}, \qquad \log \varepsilon_t^G = \rho_G \log \varepsilon_{t-1}^G + \eta_t^G, \qquad g^y = 0.2$$

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}$$
$$\bar{y}\,\hat{y}_t = \bar{c}\,\hat{c}_t + \bar{\imath}\,\hat{\imath}_t + g^y \bar{y}\,\hat{g}_t$$
$$\delta\,\hat{\imath}_t = \hat{k}_t - (1-\delta)\,\hat{k}_{t-1}$$

$$\bar{\imath} = \delta \bar{k}, \qquad \bar{y} = \bar{k}^{\alpha}$$
$$\bar{c} = (1-g^y)\,\bar{y} - \bar{\imath}$$

- Taxes finance spending with a balanced budget.
- g^y is a share of output, not a variance share.

Public spending adds a demand channel

$$Y_t = C_t + I_t + G_t, \qquad G_t = \int_0^1 T_{jt}\,\mathrm{d}j, \qquad C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^{\alpha} + (1-\delta)\,K_{jt-1}$$

$$G_t = \bar{G}\,\varepsilon_t^G, \quad \bar{G} = g^y \bar{Y}, \qquad \log \varepsilon_t^G = \rho_G \log \varepsilon_{t-1}^G + \eta_t^G, \qquad g^y = 0.2$$

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}$$
$$\bar{y}\,\hat{y}_t = \bar{c}\,\hat{c}_t + \bar{\imath}\,\hat{\imath}_t + g^y \bar{y}\,\hat{g}_t$$
$$\delta\,\hat{\imath}_t = \hat{k}_t - (1-\delta)\,\hat{k}_{t-1}$$

$$\bar{\imath} = \delta \bar{k}, \qquad \bar{y} = \bar{k}^{\alpha}$$
$$\bar{c} = (1-g^y)\,\bar{y} - \bar{\imath}$$

- Taxes finance spending with a balanced budget.
- g^y is a share of output, not a variance share.

Public spending adds a demand channel

$$Y_t = C_t + I_t + G_t, \qquad G_t = \int_0^1 T_{jt}\,\mathrm{d}j, \qquad C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^{\alpha} + (1-\delta)\,K_{jt-1}$$

$$G_t = \bar{G}\,\varepsilon_t^G, \quad \bar{G} = g^y \bar{Y}, \qquad \log \varepsilon_t^G = \rho_G \log \varepsilon_{t-1}^G + \eta_t^G, \qquad g^y = 0.2$$

$$\begin{aligned} \hat{y}_t &= \hat{a}_t + \alpha \hat{k}_{t-1} \\ \bar{y}\,\hat{y}_t &= \bar{c}\,\hat{c}_t + \bar{\imath}\,\hat{\imath}_t + g^y \bar{y}\,\hat{g}_t \\ \delta\,\hat{\imath}_t &= \hat{k}_t - (1-\delta)\,\hat{k}_{t-1} \end{aligned}$$

$$\bar{\imath} = \delta \bar{k}, \qquad \bar{y} = \bar{k}^{\alpha}$$
$$\bar{c} = (1 - g^y)\,\bar{y} - \bar{\imath}$$

- Taxes finance spending with a balanced budget.
- g^y is a share of output, not a variance share.

Public spending adds a demand channel

$$Y_t = C_t + I_t + G_t, \qquad G_t = \int_0^1 T_{jt}\,\mathrm{d}j, \qquad C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^{\alpha} + (1-\delta)\,K_{jt-1}$$

$$G_t = \bar{G}\,\varepsilon_t^G, \quad \bar{G} = g^y \bar{Y}, \qquad \log \varepsilon_t^G = \rho_G \log \varepsilon_{t-1}^G + \eta_t^G, \qquad g^y = 0.2$$

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}$$
$$\bar{y}\,\hat{y}_t = \bar{c}\,\hat{c}_t + \bar{\imath}\,\hat{\imath}_t + g^y \bar{y}\,\hat{g}_t$$
$$\delta\,\hat{\imath}_t = \hat{k}_t - (1-\delta)\,\hat{k}_{t-1}$$

$$\bar{\imath} = \delta \bar{k}, \qquad \bar{y} = \bar{k}^{\alpha}$$
$$\bar{c} = (1-g^y)\,\bar{y} - \bar{\imath}$$

- Taxes finance spending with a balanced budget.
- g^y is a share of output, not a variance share.

Public spending adds a demand channel

$$Y_t = C_t + I_t + G_t, \qquad G_t = \int_0^1 T_{jt}\,\mathrm{d}j, \qquad C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^{\alpha} + (1-\delta)\,K_{jt-1}$$

$$G_t = \bar{G}\,\varepsilon_t^G, \quad \bar{G} = g^y \bar{Y}, \qquad \log \varepsilon_t^G = \rho_G \log \varepsilon_{t-1}^G + \eta_t^G, \qquad g^y = 0.2$$

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}$$
$$\bar{y}\,\hat{y}_t = \bar{c}\,\hat{c}_t + \bar{\imath}\,\hat{\imath}_t + g^y \bar{y}\,\hat{g}_t$$
$$\delta\,\hat{\imath}_t = \hat{k}_t - (1-\delta)\,\hat{k}_{t-1}$$

$$\bar{\imath} = \delta \bar{k}, \qquad \bar{y} = \bar{k}^{\alpha}$$
$$\bar{c} = (1-g^y)\,\bar{y} - \bar{\imath}$$

- Taxes finance spending with a balanced budget.
- g^y is a share of output, not a variance share.

Six equations close the spending extension

The spending extension supplies six equations, with $g^y = 0.2$:

$$
\begin{aligned}
\hat{c}_t &= -\beta\alpha\bar{k}^{\alpha-1}\big(E_t\hat{a}_{t+1} - (1-\alpha)\hat{k}_t\big) \\
&\quad + \beta\big[\alpha\bar{k}^{\alpha-1} + 1 - \delta\big]E_t\hat{c}_{t+1}, \\
\hat{y}_t &= \hat{a}_t + \alpha\hat{k}_{t-1}, \\
\bar{y}\hat{y}_t &= \bar{c}\hat{c}_t + \bar{i}\hat{i}_t + g^y\bar{y}\hat{g}_t, \\
\delta\hat{i}_t &= \hat{k}_t - (1-\delta)\hat{k}_{t-1}, \\
\hat{a}_t &= \rho_A\hat{a}_{t-1} + \eta_t^A, \\
\hat{g}_t &= \rho_G\hat{g}_{t-1} + \eta_t^G.
\end{aligned}
$$

The two independent innovations must transmit differently to the two observables. Counting shocks is not a rank test.

Read and work: two observables

Read handout 3, §4.3, Exercise 2, *Estimating the model with output and consumption*, and complete its spending and preference-shock extensions.

- Checkpoint: two innovations must generate distinct movements in output and consumption. Count equations, then check covariance rank.
- Compare models on the same sample, observables and transformations.
- Explain which restriction each shock relaxes before interpreting a variance decomposition.

The equations just derived are the starting model, not an exercise correction.

Read and work: two observables

Read handout 3, §4.3, Exercise 2, *Estimating the model with output and consumption*, and complete its spending and preference-shock extensions.

- Checkpoint: two innovations must generate distinct movements in output and consumption. Count equations, then check covariance rank.
- Compare models on the same sample, observables and transformations.
- Explain which restriction each shock relaxes before interpreting a variance decomposition.

The equations just derived are the starting model, not an exercise correction.

Read and work: two observables

Read handout 3, §4.3, Exercise 2, *Estimating the model with output and consumption*, and complete its spending and preference-shock extensions.

- Checkpoint: two innovations must generate distinct movements in output and consumption. Count equations, then check covariance rank.
- Compare models on the same sample, observables and transformations.
- Explain which restriction each shock relaxes before interpreting a variance decomposition.

The equations just derived are the starting model, not an exercise correction.

A preference shock changes marginal utility

The alternative household objective in the handout is

$$\max_{C_{jt},K_{jt}} E_t \sum_{\tau=0}^{\infty} \beta^\tau \varepsilon^C_{t+\tau} \log C_{jt+\tau}, \qquad \log \varepsilon^C_t = \rho_C \log \varepsilon^C_{t-1} + \eta^C_t.$$

The marginal-utility channel, already shown in the previous deck, gives

$$\frac{\varepsilon^C_t}{C_t} = \beta E_t \left[\frac{\varepsilon^C_{t+1}}{C_{t+1}} \left(\alpha A_{t+1} K_t^{\alpha-1} + 1 - \delta \right) \right].$$

A spending shock enters the resource constraint. A preference shock changes the intertemporal valuation of consumption. Their observable implications must be compared, not just their names.

A preference shock changes marginal utility

The alternative household objective in the handout is

$$\max_{C_{jt},K_{jt}} E_t \sum_{\tau=0}^{\infty} \beta^{\tau} \varepsilon^C_{t+\tau} \log C_{jt+\tau}, \qquad \log \varepsilon^C_t = \rho_C \log \varepsilon^C_{t-1} + \eta^C_t.$$

The marginal-utility channel, already shown in the previous deck, gives

$$\frac{\varepsilon^C_t}{C_t} = \beta E_t \left[\frac{\varepsilon^C_{t+1}}{C_{t+1}} \left(\alpha A_{t+1} K_t^{\alpha-1} + 1 - \delta \right) \right].$$

A spending shock enters the resource constraint. A preference shock changes the intertemporal valuation of consumption. Their observable implications must be compared, not just their names.

A preference shock changes marginal utility

The alternative household objective in the handout is

$$\max_{C_{jt},K_{jt}} E_t \sum_{\tau=0}^{\infty} \beta^{\tau} \varepsilon^C_{t+\tau} \log C_{jt+\tau}, \qquad \log \varepsilon^C_t = \rho_C \log \varepsilon^C_{t-1} + \eta^C_t.$$

The marginal-utility channel, already shown in the previous deck, gives

$$\frac{\varepsilon^C_t}{C_t} = \beta E_t \left[\frac{\varepsilon^C_{t+1}}{C_{t+1}} \left(\alpha A_{t+1} K_t^{\alpha-1} + 1 - \delta \right) \right].$$

A spending shock enters the resource constraint. A preference shock changes the intertemporal valuation of consumption. Their observable implications must be compared, not just their names.

Read and work: deep parameters

Read handout 3, §4.3, Exercise 3, *Estimating deep parameters*, and work through the identification questions.

```
model(linear);
  #Kss = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
  #Css = Kss^alpha - delta*Kss;
  // model equations follow
end;
```

- Remove these derived quantities from the parameter declaration.
- Checkpoint: every candidate must update steady state and dynamics. Inspect likelihood profiles and starting values, not only t-statistics.

Read and work: deep parameters

Read handout 3, §4.3, Exercise 3, *Estimating deep parameters*, and work through the identification questions.

```
model(linear);
  #Kss = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
  #Css = Kss^alpha - delta*Kss;
  // model equations follow
end;
```

- Remove these derived quantities from the parameter declaration.
- Checkpoint: every candidate must update steady state and dynamics. Inspect likelihood profiles and starting values, not only t-statistics.

Identification is distinct from invertibility

$$\theta_1 \neq \theta_2 \implies p(Y \mid \theta_1) \neq p(Y \mid \theta_2)$$

- Without measurement errors, having at least as many independent shocks as observables is a necessary dimension check, not a sufficient rank condition.
- The forecast-error covariance must be nonsingular for the Gaussian density to exist at all.
- A nonzero interior estimate alone does not establish identification.
- Diagnostics: likelihood profiles, parameter dependence, alternative starting points.[31]

[31] Handout §4.3 exercises. A numerical optimum can sit on a flat or weakly curved likelihood.

Identification is distinct from invertibility

$$\theta_1 \neq \theta_2 \implies p(Y \mid \theta_1) \neq p(Y \mid \theta_2)$$

- Without measurement errors, having at least as many independent shocks as observables is a necessary dimension check, not a sufficient rank condition.
- The forecast-error covariance must be nonsingular for the Gaussian density to exist at all.
- A nonzero interior estimate alone does not establish identification.
- Diagnostics: likelihood profiles, parameter dependence, alternative starting points.[31]

[31] Handout §4.3 exercises. A numerical optimum can sit on a flat or weakly curved likelihood.

Identification is distinct from invertibility

$$\theta_1 \neq \theta_2 \implies p(Y \mid \theta_1) \neq p(Y \mid \theta_2)$$

- Without measurement errors, having at least as many independent shocks as observables is a necessary dimension check, not a sufficient rank condition.
- The forecast-error covariance must be nonsingular for the Gaussian density to exist at all.
- A nonzero interior estimate alone does not establish identification.
- Diagnostics: likelihood profiles, parameter dependence, alternative starting points.[31]

[31] Handout §4.3 exercises. A numerical optimum can sit on a flat or weakly curved likelihood.

Identification is distinct from invertibility

$$\theta_1 \neq \theta_2 \implies p(Y \mid \theta_1) \neq p(Y \mid \theta_2)$$

- Without measurement errors, having at least as many independent shocks as observables is a necessary dimension check, not a sufficient rank condition.
- The forecast-error covariance must be nonsingular for the Gaussian density to exist at all.
- A nonzero interior estimate alone does not establish identification.
- Diagnostics: likelihood profiles, parameter dependence, alternative starting points.[31]

[31] Handout §4.3 exercises. A numerical optimum can sit on a flat or weakly curved likelihood.

A credible likelihood is an inspectable chain

$$\text{Economic model} \longrightarrow \text{measurement} \longrightarrow (S_t, \Omega_t) \longrightarrow \ell(\theta)$$

- Check timing and state initialization before optimizing.
- Separate mechanical fit from identification and empirical relevance.
- Report transformations, sample, bounds and diagnostics.
- Takeaway: the model–observable mapping is part of the model.[32]
- Next session: combine this likelihood with explicit prior information.

[32] Handout session roadmap.

A credible likelihood is an inspectable chain

$$\text{Economic model} \longrightarrow \text{measurement} \longrightarrow (S_t, \Omega_t) \longrightarrow \ell(\theta)$$

- Check timing and state initialization before optimizing.
- Separate mechanical fit from identification and empirical relevance.
- Report transformations, sample, bounds and diagnostics.
- Takeaway: the model–observable mapping is part of the model.[32]
- Next session: combine this likelihood with explicit prior information.

[32] Handout session roadmap.

A credible likelihood is an inspectable chain

$$\text{Economic model} \longrightarrow \text{measurement} \longrightarrow (S_t, \Omega_t) \longrightarrow \ell(\theta)$$

- Check timing and state initialization before optimizing.
- Separate mechanical fit from identification and empirical relevance.
- Report transformations, sample, bounds and diagnostics.
- Takeaway: the model–observable mapping is part of the model.[32]
- Next session: combine this likelihood with explicit prior information.

[32] Handout session roadmap.

A credible likelihood is an inspectable chain

$$\text{Economic model} \longrightarrow \text{measurement} \longrightarrow (S_t, \Omega_t) \longrightarrow \ell(\theta)$$

- Check timing and state initialization before optimizing.
- Separate mechanical fit from identification and empirical relevance.
- Report transformations, sample, bounds and diagnostics.
- Takeaway: the model–observable mapping is part of the model.[32]
- Next session: combine this likelihood with explicit prior information.

[32] Handout session roadmap.

A credible likelihood is an inspectable chain

$$\text{Economic model} \longrightarrow \text{measurement} \longrightarrow (S_t, \Omega_t) \longrightarrow \ell(\theta)$$

- Check timing and state initialization before optimizing.
- Separate mechanical fit from identification and empirical relevance.
- Report transformations, sample, bounds and diagnostics.
- Takeaway: the model–observable mapping is part of the model.[32]
- Next session: combine this likelihood with explicit prior information.

[32] Handout session roadmap.